Paleomagnetism of Some Carboniferous Rocks from New South Wales and Its Relation to Geological Events

E. IRVING[1]

Dominion Observatory, Ottawa, Canada
and the Department of Geophysics, Australian National University
Canberra, Australia

Earlier results have indicated that the geomagnetic field in Australia had low or moderate inclination in Upper Silurian and Devonian time and high inclination from Permian to mid-Cretaceous time. Results reported in this paper show that moderate inclination persists to near the end of the Lower Carboniferous and that steep inclination extends down well into the Upper Carboniferous, so that most of the direction change (aside from reversals and secular variation) occurring between the Upper Silurian and mid-Cretaceous was concentrated in a comparatively short time in the Carboniferous. The inclination change corresponds to a latitude change of southeastern Australia from about 30°S to about 75°S over a period of probably not more than 20 m.y. This change coincides at its beginning with the Kanimblan orogeny (the largest tectonic event in eastern Australia) and associated marine regression, and is marked at its end by the onset of positive (reversed) polarity which persist to near the end of the Paleozoic in the Australian sections. Direction changes also occurred in Europe at this time, but they were smaller, suggesting that an episode of relative land movement (called the *mid-Carboniferous continental drift episode*) occurred before the episode invoked by Wegener and others for the late Mesozoic and Cenozoic. These results indicate that there was a correlation in the Upper Carboniferous of a change of latitude and orogeny (and possible drift episode) with a change in reversal frequency, implying a connection between events in the core and at the earth's surface. It is suggested that field reversals and these surface events have a common cause being governed by the redistribution of mass in the mantle. Other paleomagnetic evidence for 'pre-Wegenerian' drift episodes is reviewed, and it is suggested that drift might have been a recurrent process involving the repeated redistribution of continental shields and was not confined to a comparatively late stage in the earth's history.

INTRODUCTION

In the period 1955 to 1957, results [*Irving and Green,* 1958] were obtained showing the movement of the paleomagnetic pole relative to Australia during and since the late Precambrian. These pole movements were grossly inconsistent with those from Europe and North America except for recent times. This discrepancy was explained by supposing that Australia had moved relative to the northern continents. The differences were such that the results from the late Paleozoic to the present could be explained by supposing that the movement occurred predominantly in the late Mesozoic and Cenozoic, as Wegener and others have suggested. But the differences between observation from older rocks could not be explained by 'Wegenerian' drift, and Irving and Green were led to suggest that there had been an earlier drift episode in the Paleozoic. The work now described was done in order to study this problem by extending the surveys of the Carboniferous, which appears to be the critical period.

The observations were made at the Australian National University and the analysis and writing were done at the Dominion Observatory.

GENERAL PROCEDURE

For practical reasons sampling was concentrated in the Upper Carboniferous, which has many suitable exposures. Lower Carboniferous rocks are less satisfactory; there are uncertainties as to restoration of beds to their original position, and in many areas the beds have been metamorphosed in the mid-Carboniferous Kanimblan orogeny. Results from two

[1] Now at the Department of Earth Sciences, University of Leeds, U. K.

TABLE 1. Carboniferous System in New South Wales

Terrestrial		Marine	Approx. European Equivalents
Upper Kuttung	Main Glacial Stage*	Emu Creek Series	Stephanian
	Paterson toscanite*	Kullatine Series	Westphalian
	Sediments		Namurian
(Kanimblan Orogeny)			
Lower Kuttung	Volcanic Stage*	Upper Burindi	Viséan
	Basal Stage		
		Lower Burindi*	Tournaisian

* Levels sampled.

late Paleozoic (probably Permian) dikes are also recorded.

The Carboniferous System is well developed in New South Wales north of Sydney. Its main divisions are given in Table 1. The sampling sites are listed in the appendix and shown in Figure 1, where the sampling of Permian and Triassic rocks (results already published) is also given. Sampling was concentrated in three places—the lower Hunter Valley (Figure 2), Currabubulla (Figure 3), and Rocky Creek (Figure 4). At each site several oriented hand samples were taken. For igneous material, cylinders (3.5 × 3.5 cm) and for sediments, disks (3.5 × 0.7 cm) were cut from these. Disks rather than right cylinders were cut from the sediments to make it possible to look for direction trends in successive bedding levels such as may be attributed to secular variation; as it turned out, fluctuations were always random. Most natural surfaces are weathered, and samples were obtained only from roadcuts, quarries, or river beds. Orientation errors are about 3°. Sites are listed in the appendix.

Measurements were made on astatic magnetometers. In weakly magnetized material (down to 1.0×10^{-6} emu cm^{-3}) the error of individual determinations was 3°, reducing to 1° for intensities greater than 10^{-5}. In demagnetization the apparatus and methods of *Irving et al.* [1961*a*, *b*] were used. In lavas the maximum alternating field (af) strengths (1000 oe peak) available usually reduced the magnetization to a few per cent of the original, so that much of the stability range could be studied. The sediments responded less to alternating fields, but the full stability range could be studied by thermal demagnetization. Therefore, af demagnetization was generally used for lavas and thermal methods were used for sediments.

Upper Kuttung Bedded Rocks

Sampling. The Upper Kuttung series in the Hunter Valley is about 1300 m thick. It rests with angular unconformity on the Lower Kuttung and consists of conglomerates, breccias, tuffaceous sediments, and subordinate volcanics. The sedimentation is fluviatile or glacial. The beds have undergone tilting and folding which in the Hunter Valley is certainly pre-Triassic and probably Permian in age [*David,* 1950, p. 315]. About halfway through the sequence there is the Paterson toscanite, a flow about 100 m thick. The sediments from the Main Glacial Stage above the Paterson toscanite were sampled at five sites (Figure 2). The total stratigraphic thickness spanned is unlikely to be less than 150 m and is probably more than 300 m. The sites include the Paterson railway cut, where the sediments overlie the Paterson toscanite, and the classical locality at Seaham. Most of the dips are moderate but at one site (H12) the dip is almost vertical.

Upper Kuttung sediments were taken from the Currabubulla formation along Currabubulla Creek, where it is 1700 m thick (Figure 3). The beds dip to the west at about 35°. The five sites cover 450 m stratigraphically. The formation is predominantly conglomerate but some levels are fine-grained. Much of the clastic material is volcanic. Four sites are from fine-grained levels. One is a banded ignimbrite.

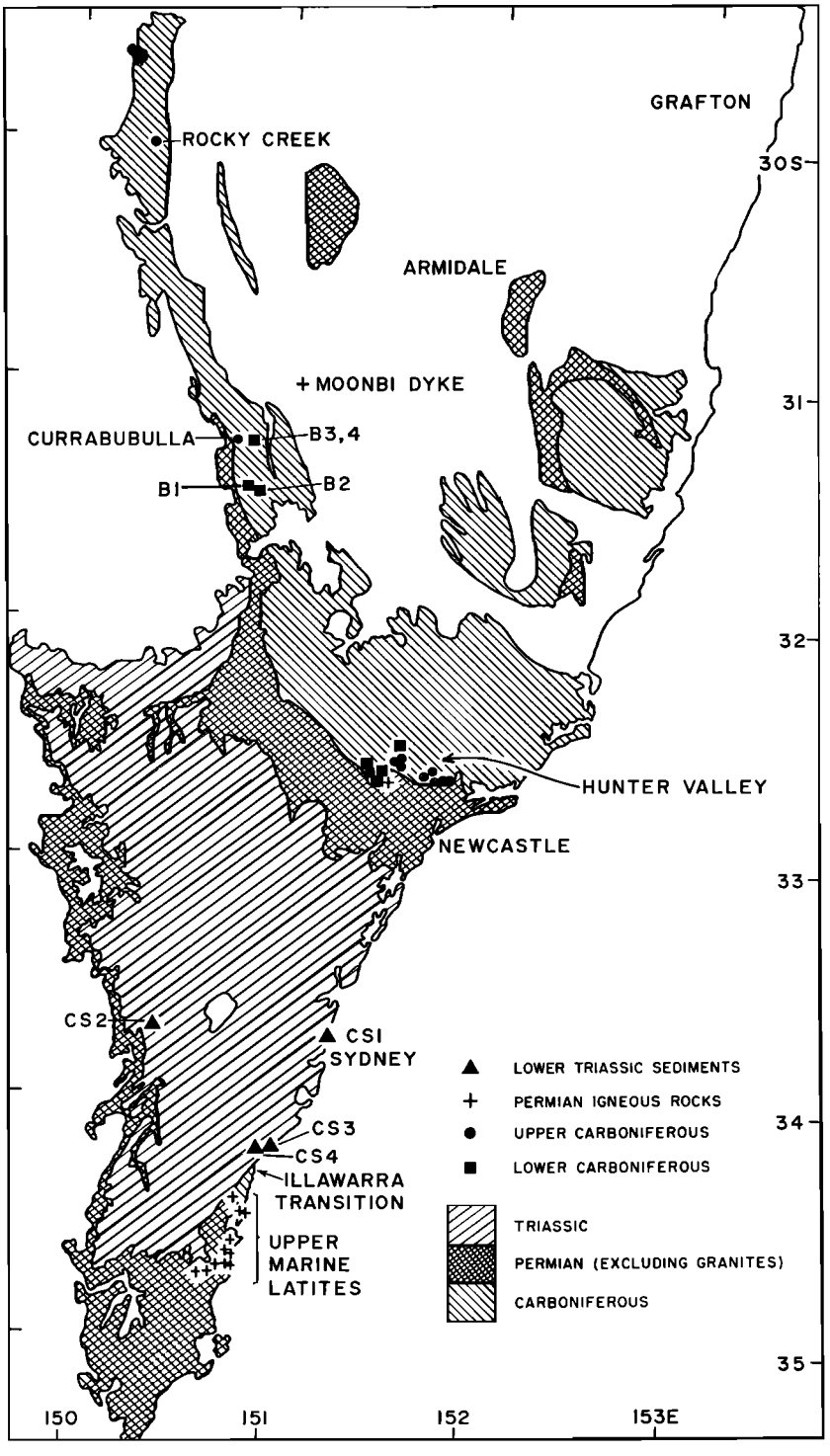


Fig. 1. Locality map.

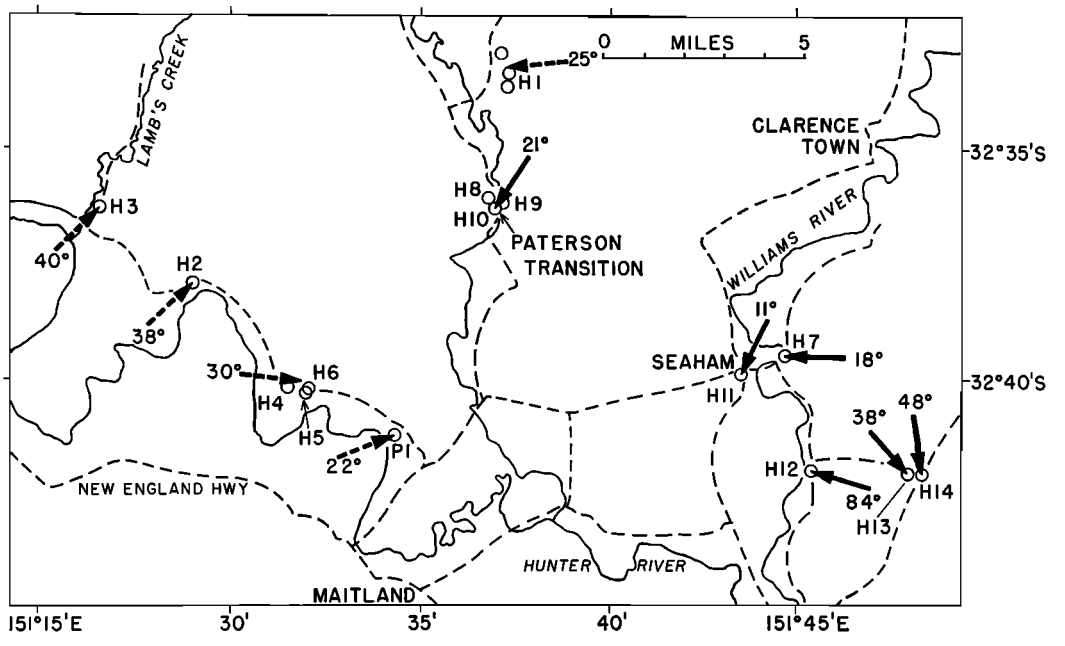


Fig. 2. Sketch-map of lower Hunter Valley showing sampling sites. Roads are shown in broken line. Full arrows give down-dip directions in sediments. Broken arrows give down-dip directions assumed in igneous rocks (see Table 2 for details).

Carey [1937] considered the formation to be predominantly glacial in origin. *Whetten* [1965] doubts this, and believes that although Carey may have been correct, origins other than glacial are possible for most of the formation; one 'varved mudstone' level (Rosedale member) he considers to be glacial. An attempt to sample the Rosedale member was unsuccessful; it is too closely fractured. Samples above and below were obtained (Figure 3).

Farther north at Rocky Creek (Figure 4) samples were taken from the Rocky Creek conglomerate. This formation contains fine-grained levels, and it is these which were usually sampled. The eight sites cover about 180 m stratigraphically and are spread over the axis of a shallow syncline. An upper limit to the age of folding is set by the overlying horizontal Jurassic sediments to the west. Together with the Currabubulla rocks they form a part of a folded strip of Carboniferous strata continuous with the Upper Kuttung of the Hunter Valley (Figure 1), and for this reason the folding is regarded as probably Permian in age (Hunter-Bowen orogeny).

The usual site statistics are given in Table 2.

Demagnetization. Test samples were demagnetized in steps up to 650°C. Four specimens from each sample were studied (orientations in the four antiparallel positions explained in *Irving et al.* [1961*a*]) to average out experimental errors.

Results from Eagleton quarry are given in Figures 5 and 6. The differences in intensity among specimens is about 10% and their decay curves follow one another closely. At first the decay is smooth, then there is an increase (peculiar to this site), and then a smooth decrease. Inclination first increases, then steadies at about 350°C, remaining little changed to 550°C (Figure 6). The within-sample scatter at Eagleton is large in the room temperatures to 350°C range (Figure 7). Between 350 and 550°C it is small, particularly in the range 400–450°C. At 600 and 650°C the directions are random and no recognizable trace of the fossil remanence remains. The dotted line above 550°C in Figure 5 is the vector mean intensity of the four specimens resolved along the stable direction present in the range 350–

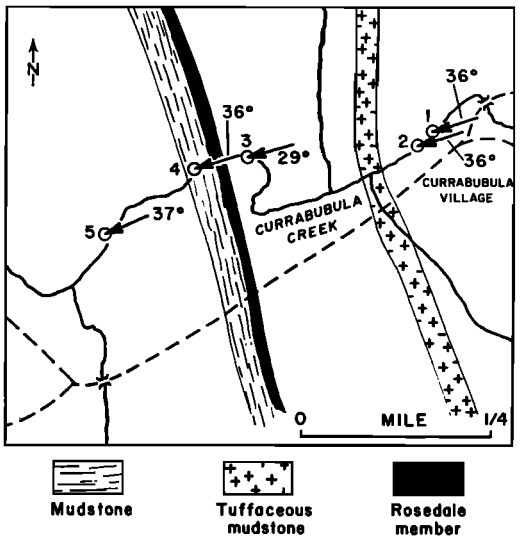


Fig. 3. Sketch-map of Currabubulla area. Roads are shown in broken line. The whole area is in Currabubulla formation. Three prominent beds are indicated [from *Whetten* 1965].

550°C. This gives a crude representation of the decay of the mean sample remanence beneath the 'noise' level for individual specimens. The directions in all specimens from Eagleton migrate away from the present field after demagnetization at 450°C (Figure 6).

Behavior of a different kind is evident at Currabubulla (Figure 8). The intensities of specimens from the same sample decay similarly, but the directions change little from room temperatures to 550°C and are in excellent agreement. Above 550°C the directions scatter (Figure 7).

Decay curves for all test samples are given in Figure 9. Hunter Valley material shows a general decrease; Rocky Creek samples increase first, then decrease; Currabubulla samples show intermediate behavior. Three demagnetization stages can be recognized. In the first stage (which normally extends from room temperature to 300 or 400°C) the decay curves may rise or fall, and the direction changes show as rotation away from the present field (Figure 6), as a decrease in scatter (Figure 7), or both. In the second stage (normally 300 or 400°C to 550°C) the intensity usually decays smoothly, the directions are well grouped, and the inclination is steeply downward. Finally, above 550°C 'noise' usually dominates; the intensity varies irregularly and the directions are random. The rotation away from the present field and decreases of scatter in the first stage are due to the removal of unstable components in the early stages of demagnetization. The consistency of directions in the second stage shows the presence of a stable component. The 'noise' in the third stage is generally a factor 10 times greater than magnetometer errors and is not due to errors in measurement. It could be an intrinsic rock property (the 'minimum intensity' of *Irving et al.* [1961*b*]) or due to errors introduced by the experiment (inadequate field cancellation, for example).

The decay curves show a wide range of blocking temperatures. The value of most interest is the maximum blocking temperature $T_{B(max)}$ which is a minimum estimate of the Curie temperature of the mineral responsible for the stable remanence. The retention of significant grouping (except at site RC3) at 550°C suggests that $T_{B(max)}$ commonly exceeds this temperature. The extended mean curve

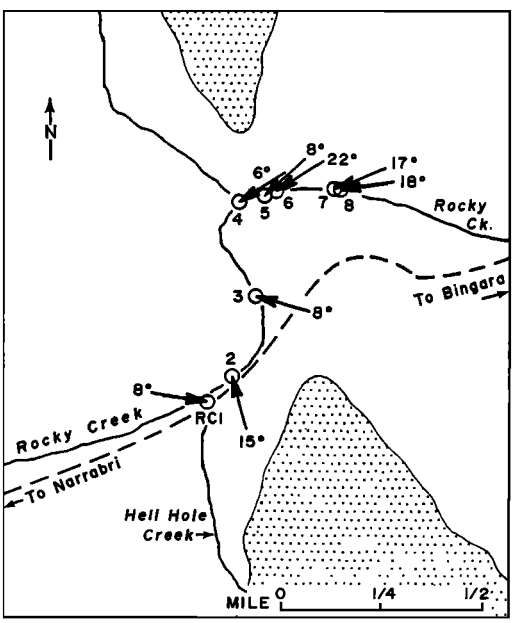


Fig. 4. Sketch-map of Rocky Creek section. Blank area is Rocky Creek conglomerate and the stippled area is the overlying Lark Hill formation. Dirt roads are shown in broken line [from *McKelvey and White*, 1964].

TABLE 2. Results from Upper Kuttung Bedded Rocks

Site	N	S	Th	Natural Remanence M	Natural Remanence D, I	Natural Remanence R_N	Cleaned Remanence T°C	Cleaned Remanence M	Cleaned Remanence D, I	Cleaned Remanence R_N	Pole
H10(S)	9	28	6	5.8	037,+46	7.56	300	6.0	156,+77	8.86	55S,170E
H11	4	18	10	9.1	022,−15	3.32	450	4.5	162,+77	3.86	56S,165E
H12	7	29	10	40.1	099,+66	6.75	450	18.5	205,+76	6.93	56S,131E
H13	5	15	12	58.4	322,−01	4.59	300	57.7	228,+83	4.67	42S,137E
H14	5	18	1	3.7	340,+60	4.00	300	4.2	194,+79	4.92	53S,143E
C1	3	13	1	150.1	288,+53	2.49	500	45.8	233,+80	2.95	41S,131E
C2	4	13	1	85.2	340,+35	2.50	500	26.0	329,+72	3.64	02S,135E
C3	3	10	1	129.9	225,+82	2.98	500	68.0	197,+75	2.97	58S,135E
C4	6	6	0.2	20.9	042,+33	5.59	500	10.7	173,+77	5.83	56S,156E
C5	5	18	1	1100.0	252,+84	4.98	500	651.0	217,+75	4.97	51S,124E
RC1	10	24	1	2.3	265,+33	9.10	400	4.2	212,+71	9.70	56S,118E
RC2	3	12	1	3.6	290,+31	2.82	400	3.9	214,+67	2.98	58S,107E
RC3	4	14	1	11.2	262,+83	3.76	300	10.9	200,+75	3.97	56S,133E
RC4	4	13	3	20.1	168,+82	3.96	300	23.5	198,+74	3.99	58S,134E
RC5	2	4	1	1.3	034,+12	1.94	400	2.2	115,+70	1.99	38S,194E
RC6	3	10	0.3	1.8	034,+22	2.09	400	2.9	166,+84	2.96	41S,154E
RC7	2	10	0.2	19.5	052,+63	1.98	400	25.0	163,+78	1.99	51S,161E
RC8	4	9	0.3	172.5	305,+50	3.73	300	178.0	269,+66	3.60	21S,109E

Notes. N is the number of oriented samples, S the number of discs, Th is the thickness in meters spanned by the samples, M is the mean intensity in emu $cm^{-3} \times 10^6$, R_N is the magnitude of the resultant giving samples unit weight and D, I its direction, T°C is the cleaning temperature used. The pole is the model A pole calculated from the direction D, I of the cleaned remanence.

(the procedure explained above may not be wholly legitimate) of Figure 5 gives $T_{B(max)} \approx$ 600°C. Perhaps the best estimate of $T_{B(max)}$ is the point at which the within-sample precision becomes random *Robertson* [1963]; at site RC3 this occurs at 445°C, and at other sites at between 575 and 645°C.

Thermal cleaning. All specimens were thermally cleaned. The cleaning temperature preferred was that at which dispersion was least, as found from pilot studies. The neck in the envelope of curves in Figure 7 suggests heating to about 450°C. Many specimens from Rocky Creek and site H13 exploded (presumably owing to zeolite transformations) at about 350°C, and heating to this temperature was not possible.

Cleaning condenses the otherwise scattered observations about a steep positive inclination (Figure 10). Examples for sites are given in Figures 6 and 11. To show the general effect, A values are calculated for each sample; A is the proportion of the total sphere of observation which 63% (one standard deviation) of the observations occupy. A varies linearly with scatter and gives a good qualitative indication. After cleaning, four internally random samples become significantly grouped and nearly half of the within-sample dispersions cover less than 1% of the sphere of observation (Figure 12). The agreement between samples from the same site (within-site dispersion) also shows a marked improvement (Figure 13); two internally random sites become significant, and 75% cover less than 2% of the sphere. This decrease in scatter is shown best in rocks of fine sandstone grade or finer which make up the bulk of the collections. A few rocks do not conform to this pattern; although they possess a stable component of magnetization, they retain their high initial dispersion after demagnetization. These rocks are all coarse-grained sandstones, and an example is detailed later.

Age of unstable components. The direction of the unstable components may be obtained from the intersection points of the great circles drawn through streaked distributions only if the differences of the bedding dips are large. This is so only for the Hunter Valley sites, and the application of Fisher's statistics to their intersections yields the mean 356, −60. The

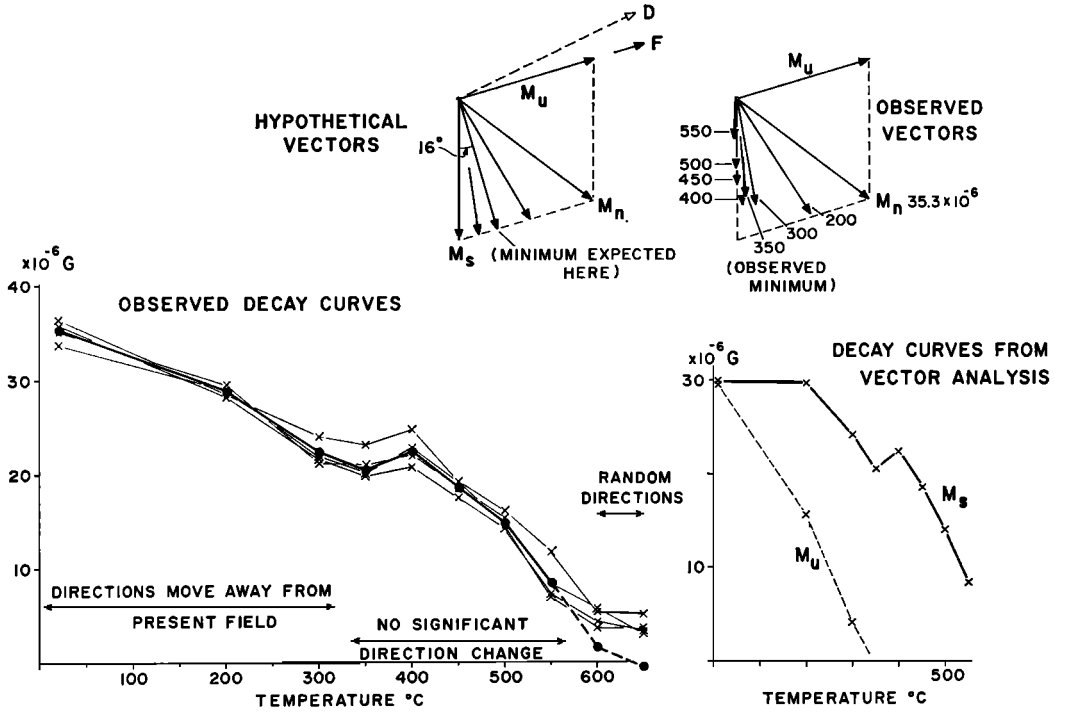


Fig. 5. Thermal demagnetization of sample K17, site H12. Intensity (emu/cm^3) decay curves are given on the left, crosses giving individual specimen values and dots the sample mean. The vector diagrams are in the plane of the remagnetization circle (Figure 6); to the right are the observed vectors, and on the left a hypothetical configuration shows the failure of vector rotation as an explanation of the minimum in the decay curve at 350°C. Below, the thermal decay of the unstable and stable components calculated by vector analysis is shown.

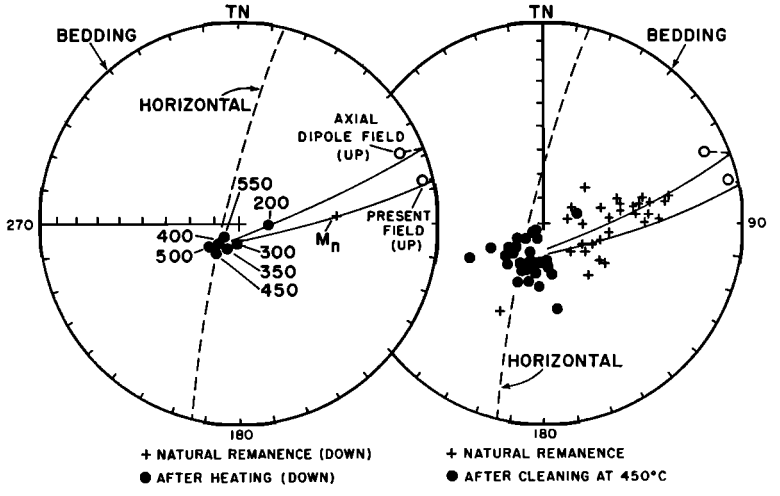


Fig. 6. Directions at site H12. On the left results from sample K17 (mean of four specimens) are given. On the right all (29) specimen direction sare plotted. Equal-area projection the primitive being the bedding plane. Present field directions (up) are marked as circles.

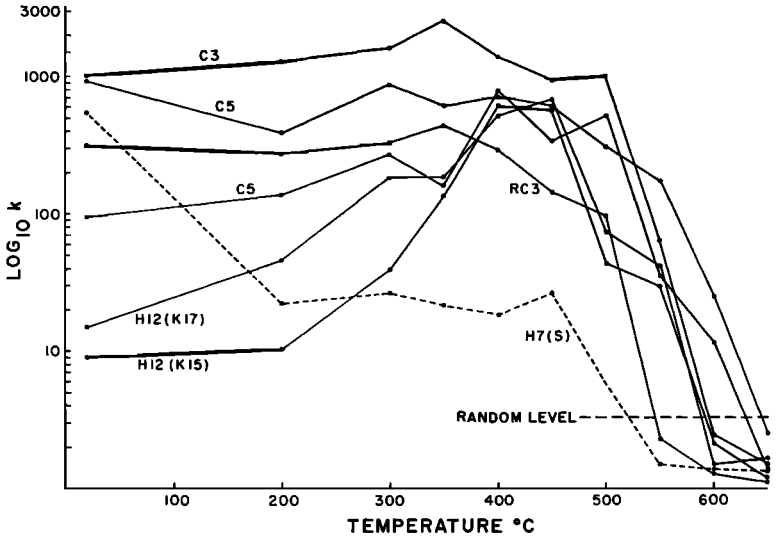


Fig. 7. Precision versus temperature. Within-sample precision k (calculated for four specimens from each sample) is plotted as a function of temperature. Random level is taken from *Vincenz and Bruckshaw* [1960].

estimate, though crude, shows that the unstable component is directed approximately along the recent earth's field, although it is not possible to say whether it is the present field or some average of the recent field, say the axial dipole field. The unstable components are therefore considered of recent origin.

Vector analysis. When rotation away from the present field is observed, the magnitudes and modes of decay of the unstable (M_u) and stable (M_s) components can be calculated. An example is given in Figure 5. M_u decays linearly, being lost below 400°C. M_s decays less rapidly, and there is a kink at 400°C which is found in all eight specimens from this site. The increase is not due to experimental errors because all curves (Figure 9) were obtained in the same run, and samples were randomly placed in the furnace. An upward kink could occur if the two components are at angles of

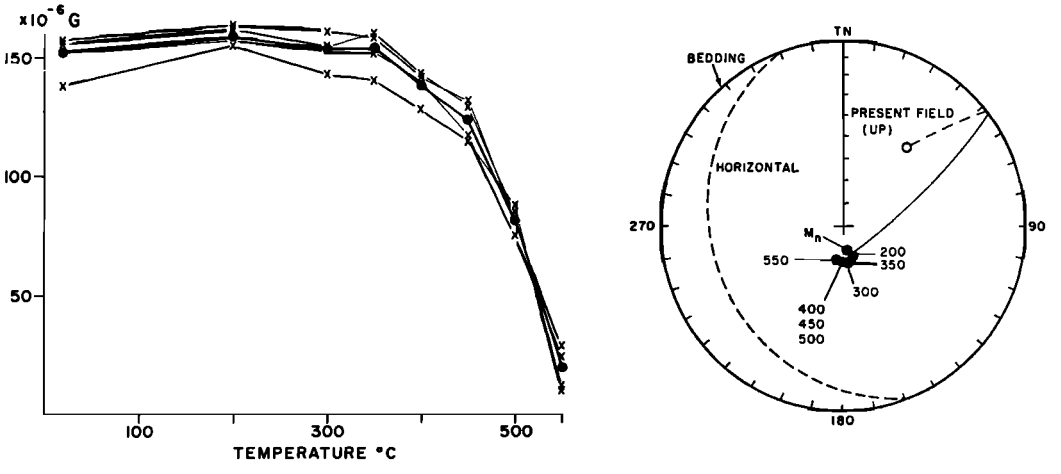


Fig. 8. Thermal demagnetization sample CB15, site C3, Currabubulla formation. Crosses are individual specimen values and dots the sample mean. Intensity in emu/cm³. The equal-area projection gives the direction changes of sample means with temperature.

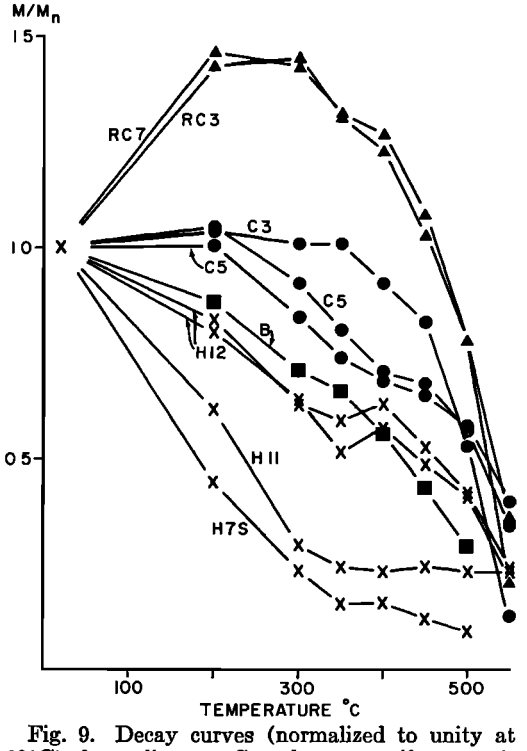


Fig. 9. Decay curves (normalized to unity at 20°C) for sediments. Sample means (four specimens each) are plotted. Triangles, Rocky Creek samples; dots, Currabubulla formation; crosses, Hunter Valley samples; squares, Burindi sediment.

greater than 90° and if M_s is not affected during the early stages of demagnetization. The possible increase from the observed configurations is only 3%, whereas the observed increase is 10%. It seems that in the interval 350–400°C either a stable antiparallel component is being removed or a new magnetic mineral coupled positively to the existing remanence is produced. Other analyses yield curves of the expected form—a rapid decay of M_u and a slower decay of M_s, curves being of convex-upward type and without kinks. M_u is sometimes negligible but is commonly about as large as M_s.

Age of stable components. The fold test of *Graham* [1949] was first applied. The initial site directions (Figure 14 and Table 3) are widely scattered for in situ orientations of the samples, and there is no significant improvement in precision after correction for tilt. *McElhinney's* [1964] tabulation shows that for the change to be significant (at $P = 0.05$) the ratio of the precisions should exceed 1.8 for $N = 18$. After cleaning, if results are plotted for the beds in situ, there is a small but significant improvement (Figure 4*b*), and after correction for tilt the directions agree very well (Figure 4*d*). Thus, in the Hunter Valley certainly, and probably also for the other localities where the tilting is probably but not certainly pre-Mesozoic, the magnetization is Paleozoic. Therefore it is Permian or Upper Carboniferous. The directions in Permian igneous rocks [*Irving and Parry*, 1963] have steep positive inclinations, and it is possible that this entire sedimentary sequence was magnetized in Permian time. The following observations were made to study this possibility.

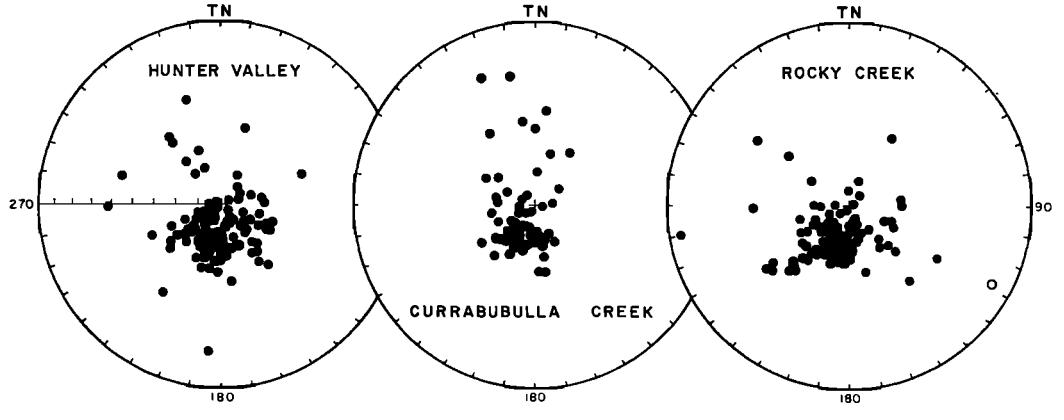


Fig. 10. All specimen directions (corrected for tilting) in Kuttung sediments. Hunter valley, 5 sites, 108 specimens. Currabubulla formation, 5 sites, 60 specimens. Rocky Creek conglomerate, 8 sites, 96 specimens. Equal-area projection.

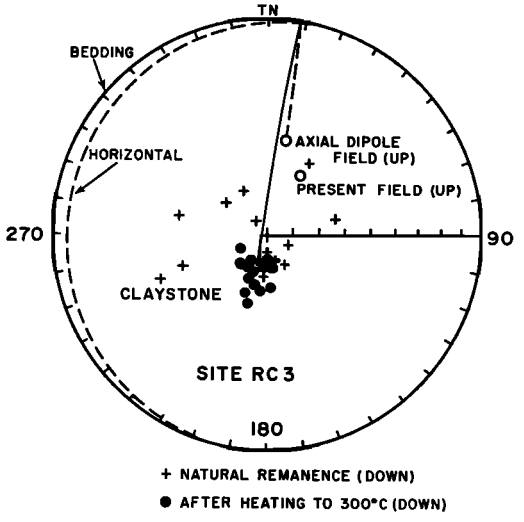


Fig. 11. Remanence directions at site RC3. Equal-area projection.

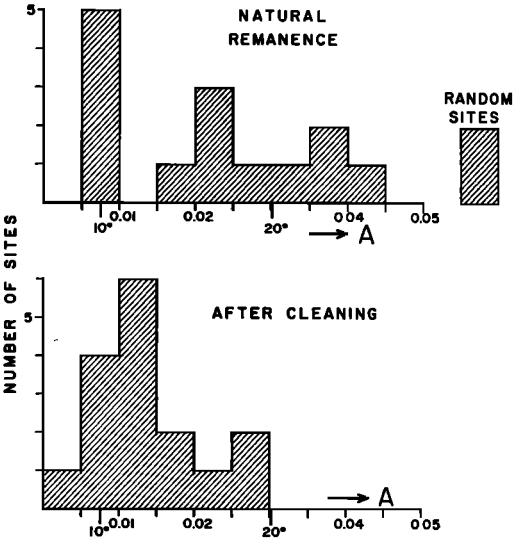


Fig. 13. Within-site dispersion. As for Figure 12.

At site RC7 a core passed through an andesite pebble. Specimens were cut from the pebble and from an adjacent claystone layer. Their decay curves (Figure 15) show the initial rise characteristic of Rocky Creek material.

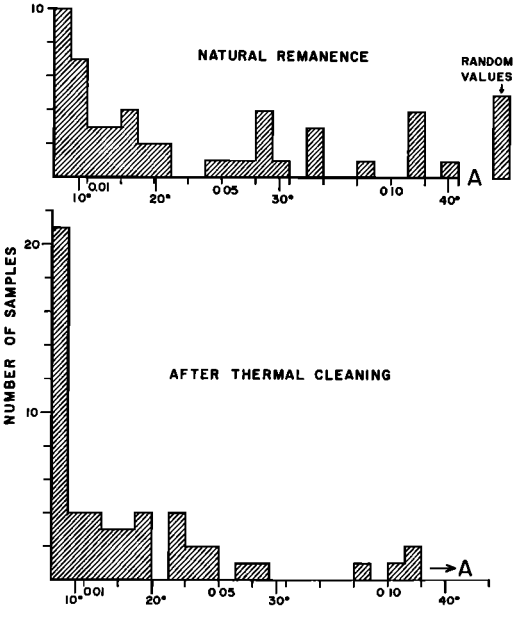


Fig. 12. Within-sample dispersion for 53 samples. A values are calculated from the relationship $A = \frac{1}{2}(1 - \cos \theta_{63})$, where θ_{63} is the circular standard deviation equal to $81k^{1/2}$, k being *Fisher's* [1953] estimate of precision.

and their similarity in form and magnitude is remarkable. It suggests that they have comparable magnetic constituents. This is consistent with the supposed glacial origin of the sediments, which are thought to have been derived by mechanical erosion from igneous rocks of which the pebble is representative. The directions in both pebble and sediment rotated away from the present field and are then consistent up to 500°C (Figure 16). The sediment has steep inclination but the pebble gave a

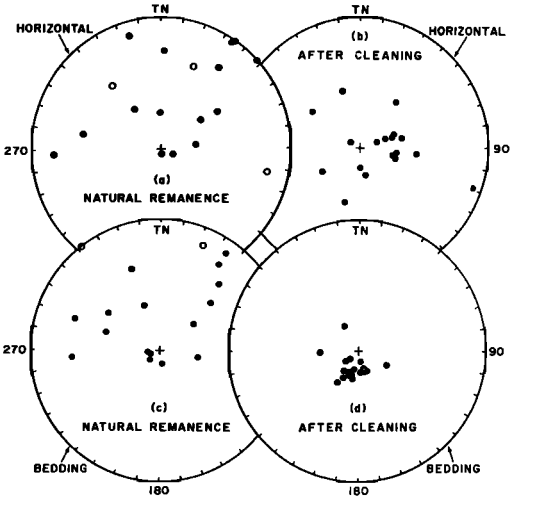


Fig. 14. Site directions in Kuttung sediments. Equal-area projection.

TABLE 3. Effect of Cleaning and Tilt Correction on Precision of 18 Site Means in Upper Kuttung Bedded Rocks

Magnetization	Reference Plane	D, I	R	k
Natural remanence	Present horizontal	018, +41	10.54	2.3
Natural remanence	Bedding	352, +60	13.11	3.5
Cleaned remanence	Present horizontal	091, +78	15.75	7.6
Cleaned remanence	Bedding	201, +79	17.64	47.6

Notes. k is Fisher's estimate of precision, R is the magnitude of the resultant, and D, I its direction.

direction 50° away. The simplest explanation is that the pebble acquired its present remanence before deposition but was too large to be oriented in the ambient field. It is unlikely that the sediment was secondarily magnetized; otherwise, the pebble, with its comparable stability characteristics, would have the same direction, which is not true. This is a variant of the conglomerate test of *Graham* [1949].

At site RC8 (Figure 17) the rock is very coarse-grained, having pebbles up to 3 cm in size, and the directions are highly scattered. Unlike the behavior of fine-grained rocks (Figures 6, 7, and 11), scatter does not diminish on demagnetization (Table 2); the directions tend to move systematically away from the present field, the mean shifting by 24°, but the scatter remains. The preservation of scatter argues against any process of extensive remagnetization; the remanence can hardly have been acquired during Permian time or it would be uniform. Therefore it is reasonable to suppose that it is detrital in origin, the poor statistical alignment caused by turbulent conditions being indicated by the coarseness of grain.

At site H10 both sediment and lava (Paterson toscanite) are exposed. The sediment has downward directions and the lava upward directions, but it is not a 180° reversal (Tables 2 and 5). It seems likely that any extensive remagnetization process affecting the sediment would also have been recorded in the lava a few feet away, but there is no sign in it of a steep, positively inclined component (see below).

Finally, *Briden* [1964] has shown from laboratory heating experiments that Currabubulla specimens (site C5) do not show the rapid build-up of viscous partial TRM characteristic of rocks possessing a stable secondary magnetization.

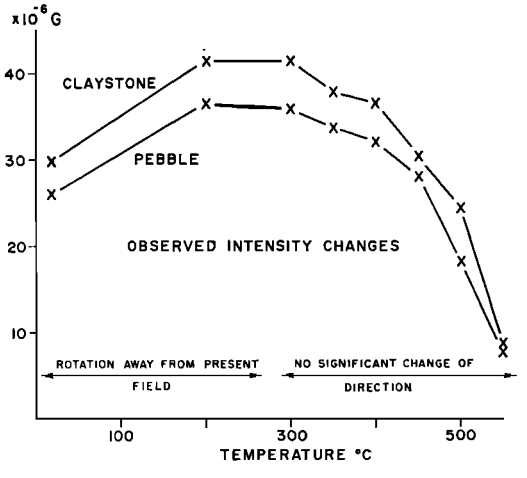


Fig. 15. Thermal decay curve (intensity emu/cm³) from site RC7.

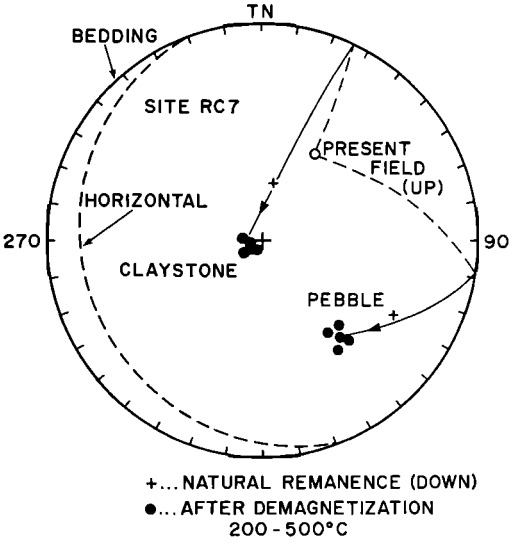


Fig. 16. Remanence directions at site RC7. Equal-area projection.

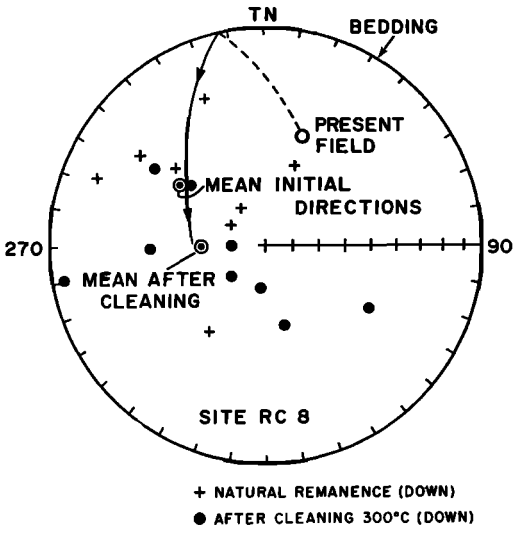


Fig. 17. Remanence directions at site RC8. Equal-area projection.

The divergence of directions of specimens from that of the present field and the laboratory studies show the high-temperature components to be stable. The tilt test shows that the stable remanence is of Paleozoic age. The three results just described suggest that it is of Carboniferous age and was acquired at or soon after deposition.

Anisotropy. Groups of specimens were heated to 600°C and cooled in the present field to induce a TRM. The declination always agrees with the field (Table 6). In igneous rocks (except for the ignimbrite, site C5) the inclinations also agree with the field. In sediments the field and remanence inclinations agree if bedding is set vertically during cooling. If the bedding is set horizontally the inclinations are about 4° less than the applied field, which shows that the direction of easiest magnetization is in the bedding plane. Artificial silts deposited in the laboratory by sedimentation through water have a preferred anisotropy in the bedding plane [*Rees*, 1961] and also have inclinations markedly less (approximately 10–20° at these inclinations) than that of the ambient field [*R. F. King*, 1955]. Because the fossil remanence in the Upper Kuttung sediments is not TRM, it is therefore possible that the observed inclinations are less than the field inclination at the time of deposition.

At Currabubulla Creek, level C5 is a banded ignimbrite, and its direction does not differ significantly from that observed in sediments (Table 2). The ignimbrite is anisotropic (Table 6), giving an inclination 3° lower than the ambient field when cooled with its banding horizontal. Therefore the possibility of a small inclination error in the sediments is not precluded, but systematic errors of 10° or more, such as King observed in the laboratory, are absent.

A more general test is to compare directions observed in the sediments with those described previously from Permian lavas which have been shown to be isotropic with respect to TRM [*Irving and Parry*, 1963]. The agreement is good (Figure 24). It could be argued that an inclination error in the sediments is present and the observed agreement arises from fortuitous field variation, but this calls for coincidences beyond what may be reasonably assumed. It is therefore believed that the inclinations in these sediments did not differ from the ambient field by more than the statistical fluctuations of the observations.

Analysis. Locality means and two-tier analysis of dispersion are given in Table 4. There is wide variation among dispersion in specimen observations (Figure 12). The dispersion between samples (Figure 13) is less, and sample means were therefore used as units. The method is not perfect, but it provides rough estimates of the dispersions present. In Table 4 the first three rows give the precision within and between sites, and the fourth row the precision within and between localities. The within-site values are in good agreement, the range in terms of circular standard deviation (θ_{63}) being 14° to 16°. Two localities have comparable between-site precisions (θ_{63} of 9° and 10°), but in the Hunter Valley it is not significant. If the between-site values provide an estimate of the magnitude of secular variation, a value of 10° (standard deviation) or less is indicated. There is not significant difference (row 4) between the means from the three localities, the direction being essentially uniform over 300 km.

Pole positions and errors calculated using the models *A* and *B* currently in use in paleomagnetic work [*Irving*, 1964, p. 49] are similar because at these high inclinations the varia-

TABLE 4. Analysis of Directions (cleaned) in Upper Kuttung Bedded Rocks
Values for the Permian Upper Marine latites [*Irving and Parry*, 1963] are added for comparison.

Site Area	D , I	R_{sites}	a_{95}	k_ω	k_b	a'_{95}	Model A Pole (error)	Model B Pole		
								Pole	R_{poles}	A_{95}
Hunter Valley, 5 sites	185,+79	4.98	6	33	*ns*	5	53S,149E(11,11)	53S,148E	4.92	11
Currabubulla, 5 sites	223,+82	4.90	13	25	63	10	42S,135E(24,25)	43S,135E	4.64	24
Rocky Creek, 8 sites	200,+78	7.80	9	30	79	7	51S,138E(17,17)	52S,138E	7.39	17
Average for Upper Kuttung sediments, 18 sites	*201,+80*	*17.66*	*5*	*45*	*ns*	*4*		*50S,140E*	*16.88*	*9*
Upper Marine Latite, 11 sites	232,+81	10.62	6				44S,132E(12,12)	46S,136E	10.01	15

Notes. R_{sites} is the magnitude of the locality resultant and D, I is direction with respect to bedding, a_{95} is Fisher's circle of confidence ($P = 0.05$), k_ω, k_b, and a'_{95} are the within-site and between-site precisions and the circle of confidence obtained from the two-tier analysis of *Watson and Irving* [1957]. R_{poles} is the magnitude of the model B pole resultant and A_{95} its circle of confidence.

tions assumed for either model are similar. The values italicized in row 4 are regarded as the best over-all summary of the data.

Basic Dikes

Moonbi lamprophyre. The dike at Moonbi Lookout intrudes granite. It is 60 cm thick. Stratigraphic evidence shows the granite to be pre-Triassic, and it is considered to be Permian like other New England granites [*Evernden and Richards*, 1962]. The dike is part of the intrusive cycle and is considered Permian (B. W. Chapell, personal communication). The fresh outcrop is very small and only three samples (nine specimens) were taken. The contacts were weathered and could not be sampled. The initial directions have low inclinations (Figure 18). After af demagnetization in 300 oe the inclinations become steep and positive. The mean direction is unchanged in 650 oe. The initial decrease in intensity is rapid, coinciding with the change in direction. Then the decrease becomes slow and linear with field (Figure 19) and the directions are vertical and constant. There is a very stable downward component. The unstable component, which dominates the NRM, is directed to the south and is not therefore due to the present field. It may have been caused by lightning.

Initially the mean direction is 187,+11 ($N = 3$, $R = 2.97$) and the mean intensity 540 $\times$ 10^{-6} emu cm^{-3}. After cleaning (450 oe), the mean direction is 253,+80 ($N = 3$, $R = 2.98$, $a_{95} = 11°$), the mean intensity 46 $\times$ 10^{-6}, and the pole 35°S, 128°E.

Dike at site H13. This basalt dike (6 samples, 16 specimens) is very different from the Moonbi lamprophyre. In one sample (PD10) the direction changes a little after af demagnetization in 20 oe. It then remains constant (to better than a degree) to 340 oe, is little changed up to 600 oe, and is changed by less

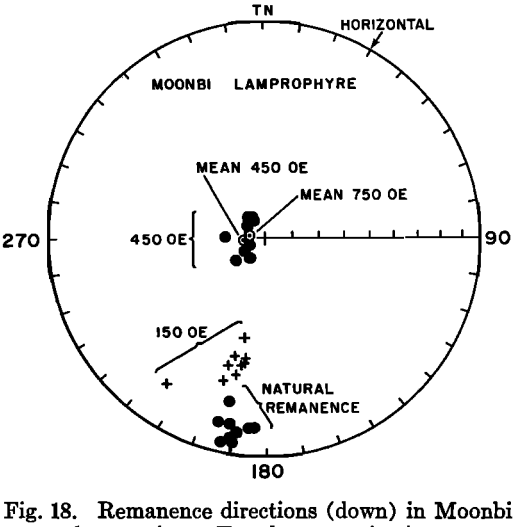


Fig. 18. Remanence directions (down) in Moonbi lamprophyre. Equal-area projection.

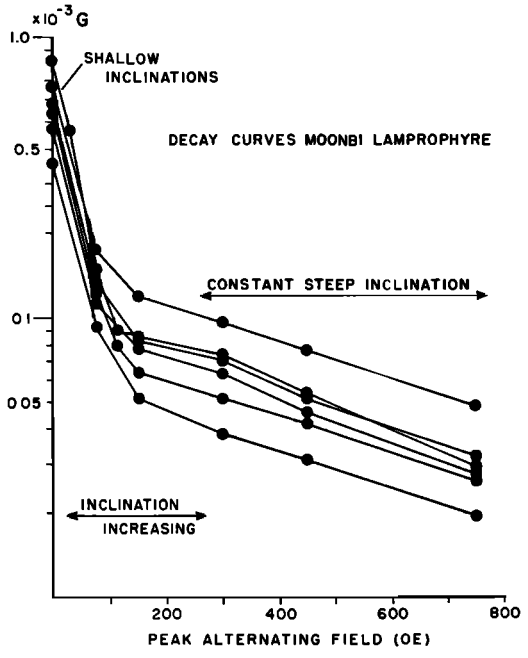


Fig. 19. Alternating field demagnetization curves for Moonbi lamprophyre.

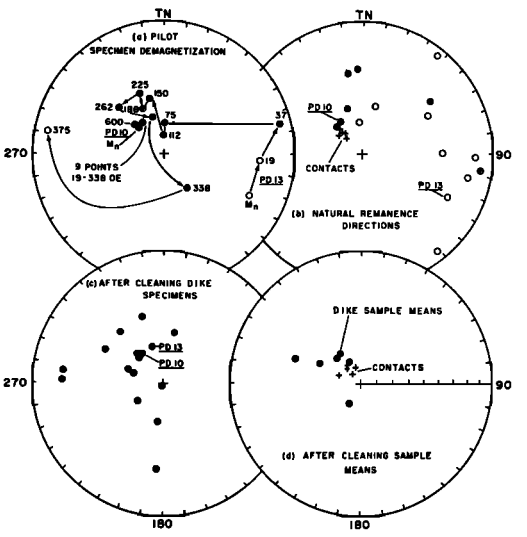


Fig. 20. Remanence directions in basalt dike, site H13, with respect to present horizontal. Contact directions (all down) are marked as crosses. Dots (down) and circles (up) are from the dike. (*a*) Direction changes in stable (PD10) and unstable (PD13) specimens. (*b*) Initial directions of all specimens. (*c*) All basalt specimens after demagnetization in 300 oe. (*d*) Sample means from basalt and contact after cleaning.

than 15° up to 750 oe (Figure 20). There is a very small initial rise in intensity and then a smooth decay to less than 2% of its original value (Figure 21). Specimen PD13 is much less stable. It moves rapidly to the same general direction as PD10 but oscillates by 10–30°. Near 300 oe the directions fluctuate widely over the sphere, and the intensities fluctuate by a factor up to 5 or more (at a level about 2% of the natural remanence). The initial directions are random at $P = 0.95$ (6 samples, $R = 3.09$). After cleaning in 300 oe they show better grouping with a mean in situ direction of 299,+69 and error $a_{95} = 14°$ ($N = 6$, $R = 5.78$, $k = 23$).

There are baked sediments at the contacts, and five samples taken within 15 cm of the dike were strongly magnetized in well-grouped initial directions (mean direction in situ 327,+72, $N = 5$, $R = 4.99$, $a_{95} = 3°$, mean intensity 408 $\times 10^{-6}$ emu cm^{-3}). The directions were little changed after thermal cleaning at 500°C (mean in situ direction 319,+78, $N = 5$, $R = 4.99$, $a_{95} = 4°$, mean intensity 286 $\times 10^{-6}$ emu cm^{-3}). The means for dike and contact rocks after cleaning agree within the statistical errors (Figure 20*d*), and the average direction of both,

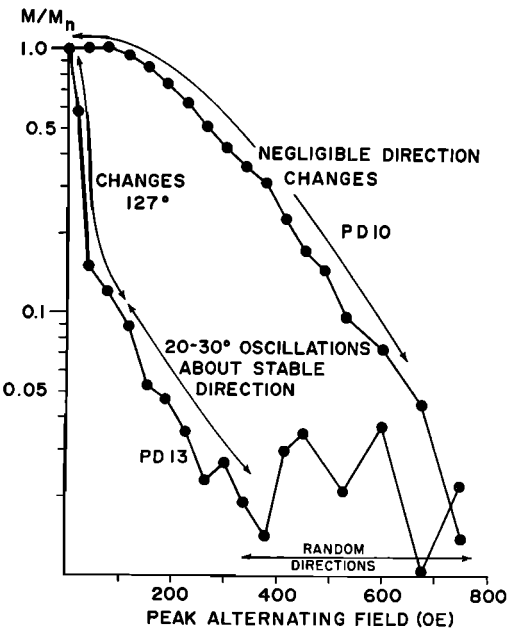


Fig. 21. Alternating field demagnetization curves for samples from basalt dike H13.

giving unit weight to each sample, is (in situ) 305,+73, $N = 11$, $R = 10.71$, $a_{95} = 8°$.

This dike was studied in an attempt to place closer limits on the age of the folding (and therefore of the magnetization of the sediments) than is possible from geological evidence. The test was not successful because of the unusually large dispersion observed. The test is now explained because it may be useful in other circumstances. The dike is considered to be of Permian age (personal communication, B. Nasher) and a feeder to the nearby basalts of the Lower Marine Series. The mean directions of stable remanence in Late Carboniferous sediments are the same as in Permian rocks (Table 4). The first possibility is that if the tilting at site H13 occurred after the deposition of the Lower Marine, the dike (being injected before tilting occurred) would be magnetized in the same direction as the sediments. If, however, tilting occurred before the Lower Marine was deposited, the dike should differ in direction from the sediments by an angle equal to the angle of tilt (38°), subject to uncertainties of order 10° owing to between-site dispersion (Table 4). The angle between the dike in situ remanence (319,+78) and the corrected sediments (228,+83) is 16°. After correction of the dike remanence for local sedimentary bedding (148,+68), the difference becomes 22°. The directions in the sediments are scattered, and the error in the mean is 20°. The observations do not provide a satisfactory test.

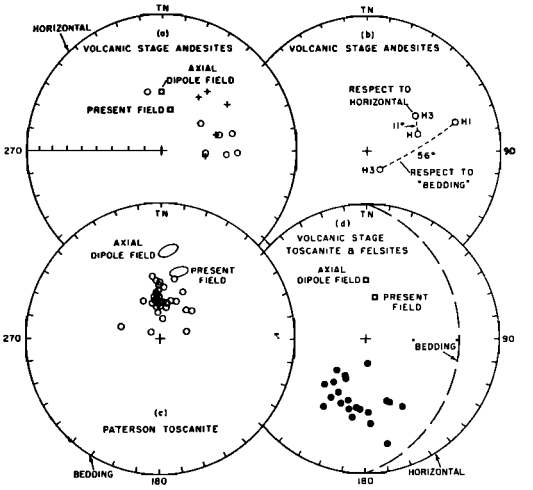


Fig. 22. Remanence directions in Carboniferous lavas. (*a*) Specimen directions (up) at site H1 (circles) and H3 (crosses) plotted with respect to present horizontal. (*b*) Means for H1 and H3 showing effect of 'bedding' correction. (*c*) Specimen directions (up) with respect to adjacent sedimentary bedding. The ovals encompass field directions which differ slightly from site to site because of variations in bedding dips. (*d*) Specimen directions (down) with respect to present horizontal. Equal-area projection.

Paterson Toscanite

The age determined geologically is Late Carboniferous (Table 1). A single K/Ar age determination for biotite from a small vein at site H8 gave 298 m.y. [*Evernden and Richards*, 1962], indicating a Late Carboniferous age on *Kulp*'s [1961] time scale. This is consistent with the age determined geologically.

It was sampled at four sites, three at Paterson (H8, H9, and H10[I]) spanning over 60 m of the flow, and one (H7) near Seaham 8 miles away (Figure 2). The directions (Figure 22*c*, Table 5) were corrected to 'bedding' obtained by averaging readings from sediments above and below the flow. The initial site directions are scattered; inclinations are negative and declinations are northerly ($N = 4$, $R = 3.86$). The precision improves after treatment in 225 oe alternating field, and the mean of sites ($N = 4$, $R = 3.99$, $D, I = 002,-67$, $a_{95} = 3°$) yields a pole at 73°S, 147°E, (model *A* errors 4, 5). The decrease in intensity after treatment in increasing alternating fields is slow and smooth (Figure 23). The difference in bedding is too small for a tilt test. For a contact test, samples were collected from a bed of red tuffaceous sandstone 30 cm beneath the base of the toscanite. They showed no obvious signs of baking, but beds closer to the contact were weathered. The initial intensities fall in the range 7 to 38 $\times 10^{-6}$ emu cm^{-3} and the directions are scattered, the mean in situ direction being close to the present field. Upon thermal demagnetization the intensities fall rapidly (Figure 9) and the dispersion increases (Figure 7), remaining scattered about the present field direction and then becoming random at about 500°C. The remanence contains no stable component and is probably a viscous magnetization acquired recently. No information is provided about the stability of the overlying toscanite.

TABLE 5. Kuttung Lavas

Site	N	S	Th	Natural Remanence M	D, I	R_s	Cleaned Remanence H_{ac}*	M	D, I	R_s
Lower Kuttung andesites (with respect to horizontal)										
H1	7	7	30	163	000,−58	6.43	225	41	072,−57	6.54
H2	9	9	7	203	315,−34	6.15	225	26	280,+63	5.22
H3	5	5	2	893	165,−17	3.87	225	53	054,−53	4.79
Lower Kuttung toscanites and felsites (with respect to horizontal)										
H4	3	7	3	50	207,+14	2.33	300	25	185,+45	2.90
H5	2	6	3	24	185,+40	1.99	300	22	174,+47	1.96
H6	3	8	2	41	220,+55	2.79	300	39	205,+55	2.97
Paterson toscanite (with respect to bedding in adjacent sediments)										
H7(L)	4	11	2	172	017,−38	3.22	225	97	003,−65	3.97
H8	3	8	4	936	357,−63	2.92	225	496	358,−66	2.96
H9	2	6	1	1301	359,−76	1.99	225	621	355,−69	1.99
H10(L)	3	8	4	466	006,−71	2.93	225	340	012,−65	2.98

* H_{ac} is peak af in oersteds used; otherwise symbols are as for Table 2.

Lower Kuttung Lavas

The Lower Kuttung Series in the Hunter Valley is divided into the *Basal Stage* (850 m), consisting of very coarse conglomerates and tuffs unsuitable for this work, and the *Volcanic Stage* (1000 m), composed chiefly of lavas (Table 1). Individual flows are few (about eight in the Gosforth section of *Browne* [1927]) and vary from 20 to 300 m in thickness. The Volcanic Stage is regarded by David and Browne as Viséan. K-Ar ages [*Evernden and Richards*, 1962] of the lavas gave a mean (five values) of 300 m.y. The radiometric age is broadly consistent with the geological age.

Andesites constitute the lower two-thirds of the sequence and were sampled at seven places (Figure 2). Sites H2 and H3 are approximately 550 m apart stratigraphically. The remanence is of unstable type. The initial directions are widely scattered (Table 5), and the intensity falls rapidly in alternating magnetic fields (Figure 23). At H1 and H3, the directions after treatment in 225 oe become grouped in a northeasterly direction (Figure 22*a*). At H2 this treatment increases the scatter, so that it is almost random. The mean in situ directions of H1 and H3 are not very different (their mean is 062,−56, $N = 2$, $R = 1.99$), whereas if corrected for dips in adjacent sediments they do differ (Figure 22*b*). The remanence appears to have been acquired after tilting. Its direction cannot be matched by any observed in younger rocks in Australia and its origin is obscure. It is possible that the 'folding' of lavas is ap-

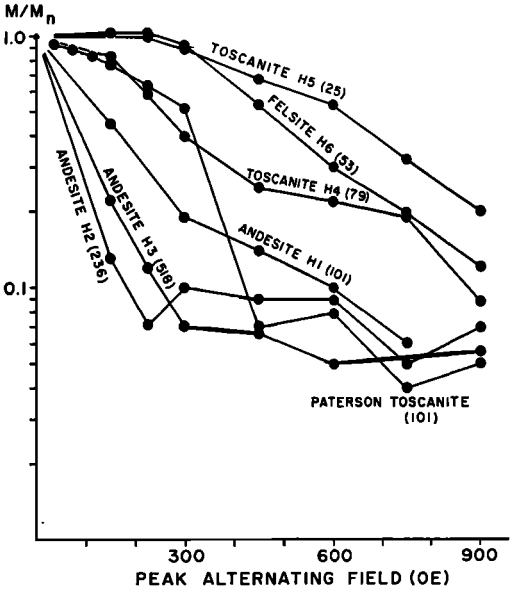


Fig. 23. Alternating field demagnetization (normalized) curves for Carboniferous igneous rocks. Initial intensities (× 10^{-6} emu/cm^3 given in brackets.

TABLE 6. TRM Directions Acquired in the Present Earth's Field (000, −65)

Sample Group	N	D, I	k	$a_{95}°$
Upper Kuttung sediments bedding horizontal	11	000, −61	545	2°
Upper Kuttung sediments bedding vertical	3	000, −65	5804	2°
Basalt dike site H13(D)	3	002, −66	380	2°
Paterson toscanite	7	359, −65	548	3°
Felsite site H6	6	358, −64	523	3°
Ignimbrite, site C5, banding horizontal	1	358, −62	...	...
Ignimbrite, site C5, banding vertical	1	000, −65	...	...

Notes. N number of samples, D, I the mean direction, k Fisher's precision, $a_{95}°$ the circle of confidence ($P = 0.05$).

parent and that they were laid down at their present relative angles. The field evidence presented by *Browne* [1927] makes this possibility very unlikely.

The overlying toscanites and felsite were sampled at sites H4, H5, and H6, which are spaced over a distance of a mile along the road from Rosebrook to Maitland (Figure 2). Their stratigraphical spacing is 160 m and 30 m. Their magnetization is very hard; the intensity falls slowly and smoothly in increasing alternating magnetic fields (Figure 23), and in some specimens the directions are little changed after treatment in a field of 900 oe. The mean directions (Figure 22*d*) relative to the present horizontal for these three flows are 187,+50 ($N = 3$, $R = 2.96$), and the pole is at 67°S, 110°W. If correction for an easterly tilt of 30° as shown in *Browne's* [1927] map is made, the mean direction becomes 154,+45 and the pole 84°S, 46°E.

Lower Burindi Sediments

Samples were collected from the Parry group [*Crook*, 1961*b*]. The sedimentation is marine and fluviatile. The beds were folded and slightly altered during the late Paleozoic. The group comprises a lower part (the Baldwin formation, not sampled) of Upper Devonian age and an upper part (the Goonoo Goonoo mudstone) which is Tournaisian and is overlain by beds of Lower Kuttung age. Samples were collected from four sites (Figure 1, Table 2) in the Goonoo Goonoo mudstone, which has a maximum thickness of over 5000 m and consists of mudstone with conglomerates and sandstones.

Samples from two sites (B3 and B4) were too weak to be measured. Specimens from site B1 were very strong, ranging from 94 to 194 $\times 10^{-6}$ emu cm^{-3}. The decay curve at site B1 falls smoothly (Figure 9) and the directions remain little changed up to 500°C. At B2 the initial directions were scattered, and this scatter is reduced by demagnetization. After cleaning at 450°C the directions with respect to bedding are as follows: site B1, 2 samples, 8 specimens, 150,+85, $R = 1.98$; site B2, 2 samples, 12 specimens, 109,+74, $R = 1.99$. The angle between them is 12°. In situ these directions are 101,+69 for B1 and 103,+74 for B2; the difference is 6°. The increase in the difference after correction for tilt is not significant at the present level of sampling, but it suggests that these beds were magnetized after folding, perhaps in the late Carboniferous or Permian during deep burial. No information is provided about the Carboniferous field.

Discussion

Kiaman magnetic interval. The downward inclinations in the late Carboniferous pass unchanged into the Permian (Figure 24).

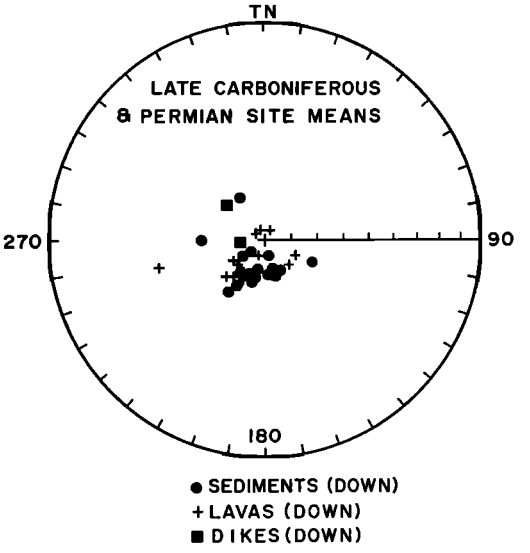


Fig. 24. Site directions in Kiaman rocks. Lava results are those of *Irving and Parry* [1963]. Equal-area projection.

Stratigraphically they extend from the Paterson transition at site H10 to the Illawarra transition between Upper Permian lavas and Lower Triassic sediments (Figure 1). It has been suggested [*Irving and Parry,* 1963] that during this time (referred to as the *Kiaman magnetic interval,* named for *Mercanton's* [1926] initial discovery at Kiama, N.S.W.) the earth's field had constant positive (or reversed) polarity, as shown by the remanence directions. This suggestion was based not only on the Australian observations (which cover only a fraction of the total time) but also on the fact that 23 rock units from other continents were also positively magnetized; it was argued that, since the units were distributed over the earth through this time span, the chance of negative (or normal) polarity occurring is small. This has since proved so.[2]

Better definition of the limits of the Kiaman magnetic interval are needed, but present data place the lower boundary in the middle of the Upper Carboniferous and the upper boundary in the Upper Permian.

The declination and inclination of the dipole field in Australia in the Kiaman are given in Figure 25. Such maps are useful for predicting results from new test localities. If discrepancies occur, relative movement (a possible cause) can be obtained directly; for example, should the test locality have undergone rotation φ, then φ is simply the difference between the test locality declination and that on the map.

Field axis in late Paleozoic and Mesozoic time. Earlier data [*Irving and Green,* 1958; *Irving,* 1964] suggested that Kiaman and Mesozoic poles differed. As more results have been accumulated and systematic demagnetization completed, the poles (solid dots and squares in Figure 26) have coalesced into a single coherent group. The scatter of observations (Figure 27) is small and is attributable to statistical errors (3–13°, average about 7°). The higher polar scatter arises from the difficulty of estimating pole positions from high-latitude observing stations (see equations for calculating poles in *Creer et al.* [1957]). For a given polar error, the direction error at an observing locality near the pole has to be about 3 times (depending a little on the model assumed) that observed at the equator—about 9 times more samples for a given dispersion are needed. This difficulty may be partly offset by field dispersion being less for high than for low latitudes [*Creer,* 1962; *Cox,* 1962]; but it is not known whether such an effect is present in the ancient field, and it is certainly not evident in the experimental data for the Australian late Paleozoic and Mesozoic in which the within-site dispersion greatly exceeds that between sites (Table 4, and *Watson and Irving* [1957]). Limitations are set by geology; even under good observing conditions (abundant exposure and high magnetic stability) in the Tasmanian dolerite (9.30, Figure 26) the polar error is still 6° [*Irving,* 1963]. Other poles have errors ranging from 9° to 25°, averaging about 14°. For a period of time spanning about 200 m.y. from the Main Glacial Stage in the Late Carboniferous to the middle Cretaceous (poles 10.09 and 10.10, Figure 26), the geomagnetic field has not changed its axis (it may have reversed its polarity) relative to southeastern Australia within the accuracy of present observations. This destroys the experimental basis for the reconstruction of the past longitudinal position of Australia relative to Europe [*Irving,* 1964, p. 270]. The mean of Kiaman and Mesozoic poles is 48°S, 142°E ($N = 15$, $R = 14.8$, $K = 61$, $A_{95} = 5°$).

Carboniferous transition. Irrespective of sign

[2] There is now a total of 41 late Carboniferous and Permian rock units with positive polarity. There are also two occurrences of negative directions. First, an unspecified number of samples from the Lower Permian Yuzagol suite of the U.S.S.R. are reported negative [*Vlassov and Popova,* 1963]; the authors say that 'a few of the Yuzagol samples were found to be remagnetized in the direction of the present geomagnetic field,' but no stability studies are reported. Second, *Helsley* [1965] found 3 negative samples in a collection of about 100 reversed ones from the Lower Permian Dunkard Series of the United States. He states that his result 'should be regarded as preliminary.' Demagnetization work is reported but no studies of the *full* stability range. It is common experience in paleomagnetic work to find (e.g., *Mumme* [1962] or site H11 in the work (Table 2)) a positive polarity marked by a secondary negative component, or vice versa [e.g., *Chamalaun and Creer,* 1964]. Therefore, until their stability characteristics are known, these records of negative polarity are not regarded as evidence against the hypothesis of a consistently reversed field in late Paleozoic time.

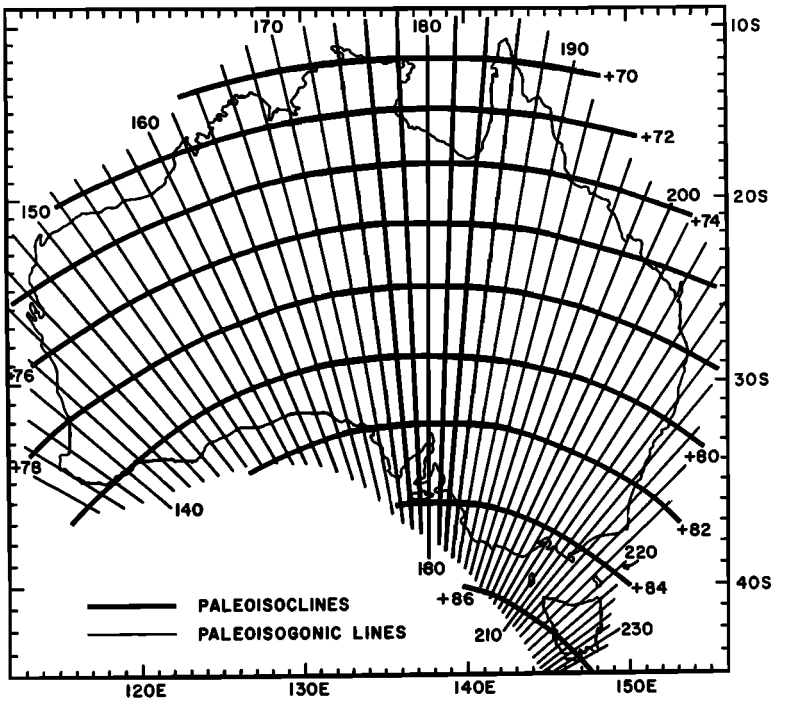


Fig. 25. Dipole field for Kiaman magnetic interval.

(positive and negative, respectively), the mean direction of the toscanites and felsites of the Volcanic Stage of the Lower Kuttung agrees remarkably well with that from the Mugga porphyry, which is Upper Silurian or Lower Devonian [*Briden,* 1964; early work by *Green,* 1959]. Both are based on demagnetization studies. From the intervening Devonian, early results without demagnetization are more scattered but have the same general direction [*Green,* 1961]. The scatter may be due to partial instability. The results suggest a rapid polar movement relative to Australia, an intermediate direction being recorded in the stratigraphically intermediate Paterson toscanite. The simplest path is plotted in Figure 28. Because of reversals, more complicated paths could be constructed by connecting antipoles. The polar change occurred in a time short compared with the stable field configurations before and afterward.

An early attempt, based on six samples, to time these polar changes using the K/Ar methods [*Evernden and Richards,* 1962] shows no significant age difference between the Volcanic Stage lavas and the Paterson toscanite, suggesting either that the difference is beyond the resolution of the method or that the former have suffered partial Ar loss during the Kanimblan orogeny. According to the correlations of *David* [1950], the polar change occupies all or part of the Namurian and Westphalian time (Table 1). Taking their duration as 20 m.y. (*Kulp's* [1961] time scale), we estimate a rate of change of about 2° per million years.

The latitude change indicated paleomagnetically is in remarkable agreement with the paleoclimatic evidence. The Lower Carboniferous has coral reefs and oölitic limestones. Reefs are unknown from the Upper Carboniferous until the middle Tertiary [*Teichert,* 1952]. The reef 'cut-off' coincides with the latitude transition. The late Carboniferous and Permian have deposits which *observers* agree are partly glacial in origin; the results herein are from the glacial sequences themselves, and they indicate high latitude.

One line of evidence favoring continental drift is the occurrence of late Paleozoic glacial beds in areas spanning half the globe. It is argued that the area becomes manageable climatically only if these continents were together

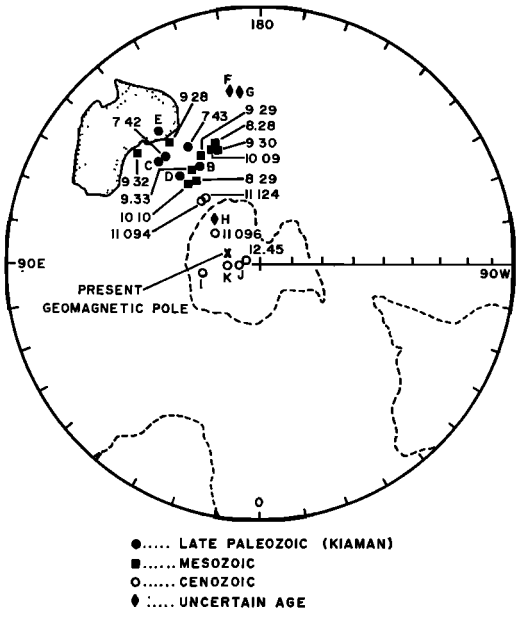


Fig. 26. Paleomagnetic poles from late Carboniferous and younger rocks in Australia. Index numbers refer to the list of *Irving* [1964]. Later results not recorded in that list are *B*, Main Glacial Stage, Hunter Valley (herein); *C*, Currabubulla formation (herein); *D*, Rocky Creek conglomerate (herein); *E*, Moonbi lamprophyre (herein); *F*, Milton monzonite [*Robertson* 1964]; *G*, Luddenham dike [*Manwaring*, 1963]; *H*, Peat's Ridge basalt [*Manwaring*, 1963]; *I*, Queensland Tertiary intrusives [*Robertson*, 1962]; *J*, Queensland Tertiary intrusives [*Robertson*, 1962]; *K*, Gayndah basalt, Queensland [*Robertson*, 1962]. Equal-area projection.

near the south pole at that time. In fact the glaciated area is still large (93° of arc on *Du Toit's* [1937] reconstruction). To explain this, *Köppen and Wegener* [1924] suggest a polar shift from southern Africa toward Australia between Carboniferous and Permian times. Du Toit followed their plan, and *L. C. King* [1962] has greatly elaborated it, basing his arguments on the general eastward migration with time of glacial and other facies. He says that there is 'from early Carboniferous until the mid-Permian, a clear progressive movement, first of glacial phenomena and then of cold-temperate coal measure accumulation from west to east Gondwanaland.' For a comparison the Australian poles are plotted on his map (Figure 28). King's Upper Carboniferous pole (estimated climatically) corresponds to the Lower Carboniferous pole determined magnetically, and his Permian pole is only about half as far toward Australia as the Australian paleomagnetic pole. The form of the paleomagnetic transition across Gondwanaland is similar to, but larger than, that suggested by King, but the timing is different—the paleomagnetic change occurs within the Carboniferous, whereas King suggests motion between Carboniferous and Permian times.

It is interesting to try to resolve this conflict. The paleomagnetic picture suggested is that the west Gondwanaland (Africa and South America) glacials are predominantly Devonian or Lower Carboniferous, and the east Gondwanaland (India and Australia) glacials are Lower Carboniferous or Permian. This is consistent with the dating of the east Gondwanaland glacial beds. It is also consistent with the dating of the west Gondwanaland beds (there are Devonian glacials in South Africa, and Silurian, Devonian, and Lower Carboniferous glacials in South America) except for the

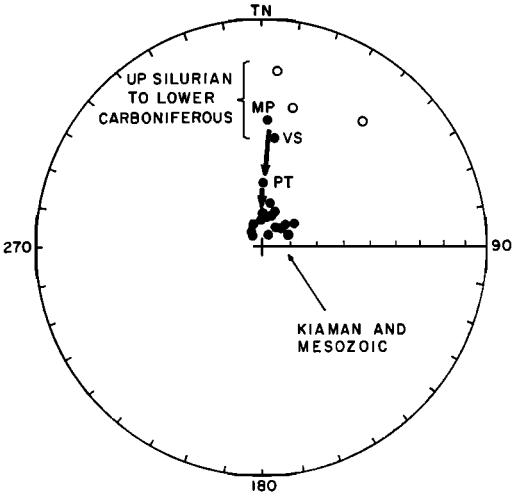


Fig. 27. Mean directions in Australian Upper Paleozoic and Mesozoic units. All directions are on the upper hemisphere (irrespective of sign) of an equal-area projection. The dots show results based on demagnetization studies, and circles show early data based on observation of NRM which require reinvestigation. There are 15 Kiaman and Mesozoic rock units. *PT*, Paterson toscanite; *MP*, Mugga porphyry (Upper Silurian or Lower Devonian); *VS*, Volcanic Stage lavas—toscanite and felsites of the Lower Kuttung. The arrow gives direction of field change as determined by superimposition of beds.

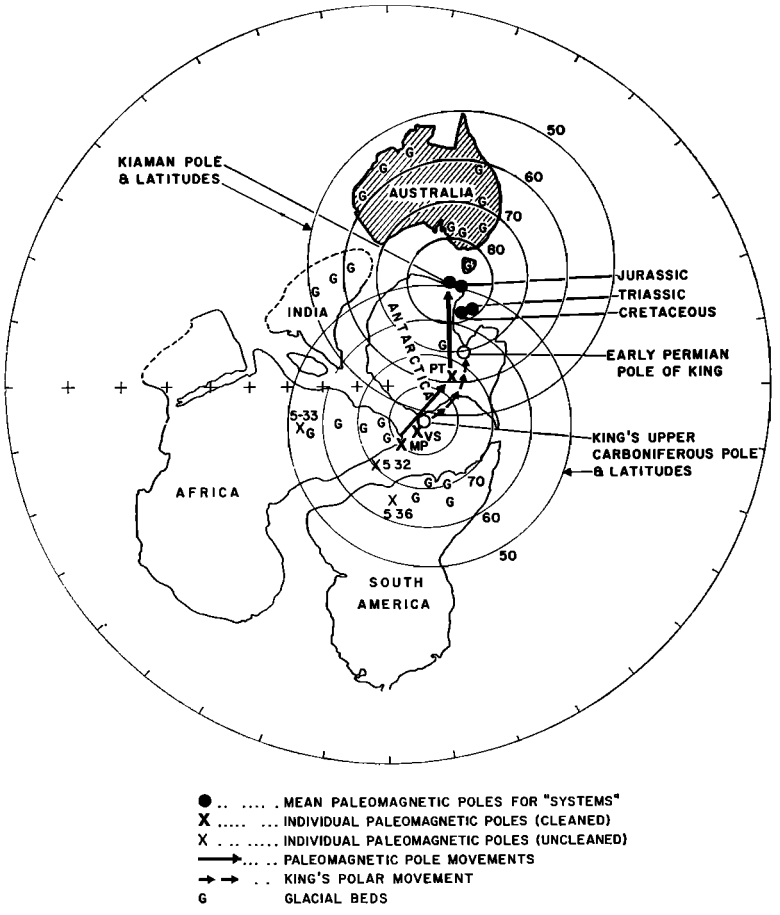


Fig. 28. Upper Paleozoic pole movement relative to Gondwanaland. The outline is *L. C. King*'s [1962] map for the Upper Carboniferous. Latitudes for the Upper Carboniferous by L. C. King, and for the Kiaman (late Carboniferous and Permian), determined paleomagnetically, are given. The poles are from Australia, lettered poles as in Figure 27, and numbered poles (Devonian) as in the list of *Irving* [1964].

Dwyka Series, which is commonly [*L. C. King*, 1962; *Furon*, 1963] regarded as Upper Carboniferous; it is this designation of the Dwyka as Upper Carboniferous which conflicts with the paleomagnetic evidence and which appears to govern, to a large extent, King's dating of his polar change. However, we note that *Haughton* [1963, p. 199] avoids assigning the Dwyka to any particular part of the Carboniferous, which indicates that there is some uncertainty as to its age. It is therefore possible that the paleomagnetic polar transition and the polar movement suggested by King and earlier workers are manifestations of the same event and that the timing conflict arises from uncertainties in the assigned ages of glacial beds. Also, we need to know whether the polar change is recorded paleomagnetically in the Carboniferous of the other southern continents. There are data only from South America, where *Creer* [1965] records a fast change in the Upper Paleozoic. There are large uncertainties in the dating of beds, however, and a comparison with the Australian transition is not attempted.

Hypothesis of a mid-Carboniferous drift episode. In Europe a polar movement of about 20° is indicated between Early Carboniferous (e.g., *Wilson and Everitt*, 1963] and Late Carboniferous and Permian times (e.g., *Creer et al.*, 1959; *Everdingen*, 1960]. The Australian and

European polar changes have the same direction (this can be seen in *Irving* [1964, Figure 6.11]), so that half of the Australian polar change can be attributed to polar wandering and half (20°) to movement of Australia relative to Europe (continental drift). The latter will be called the *mid-Carboniferous continental drift episode.*

Assuming that the Australian data record the movement of Gondwanaland as a whole, a 40° northward movement is indicated. The changes began near the end of Lower Kuttung time, and this 'was marked by the great Kanimblan epoch of orogeny and plutonic intrusion, the greatest and most important in the history of the Tasman geosyncline' [*David,* 1950, p. 325]. They correspond in Europe to the acme of the Hercynian orogeny. This drift episode appears therefore to have a similar time relationship to Hercynian-Appalachian orogeny as the Upper Cretaceous–Tertiary drift event does to the Laramide-Alpine orogeny.

Hypothesis of recurrent drift. If the suggestion made above is correct, it is not unreasonable to suppose that there have been numerous drift episodes spread through geological time. On Du Toit's ideas, Asia is composed of fragments from both Laurasia and Gondwanaland, and there is paleomagnetic evidence for the Tertiary welding of India to Asia [*Clegg et al.,* 1956; *Deutsch,* 1958]. It is therefore possible that Laurasia and Gondwanaland themselves were made up of crustal fragments brought together by earlier drift episodes. Already there are paleomagnetic results which suggest that this might be so, as *Khramov et al.* [1965] have recently suggested. The evidence is seen most generally in the analysis of polar scatter within and between continents [*Irving,* 1964, Figure 6.6]. From Permian time to present the agreement among poles from the same present-day continental region is reasonably good; the angular difference among poles in 80 or 90% of cases is less than 20°, much of which can be ascribed to experimental error or to second-order departures from a dipole field. For Carboniferous and earlier times, however, the within-continent agreement is poor and most poles differ from one another by much more than 20°. I have argued that the increase of within-continent polar scatter may be due to field variations in the long time spans involved and to the inadequacy of the observations, but this type of criticism is becoming less supportable as new results appear. For example, Devonian poles from studies of rocks from western Europe [*Stubbs,* 1958] are about 30° south of those from the Siberian Platform [*Vlassov et al.,* 1961]. The Lower Carboniferous pole from rocks of western Europe [*Wilson and Everitt,* 1963] are a similar distance south and east of those from the Siberian Platform [*Vlassov and Kovalenko,* 1961]. Yet the Permian rocks from the two regions [cf. *Irving,* 1964, Figure 6.10] are in good agreement. Farther east, discrepancies of comparable magnitudes are appearing between Triassic and Permian poles from eastern Asia [*Wang et al.,* 1960; *Vlassov and Popova,* 1963], Europe, and the Siberian Platform. Taken at their face value these data suggest that the Siberian Platform relative to western Europe lay about 30° to the south and east of its present position in Devonian and Lower Carboniferous times. They came together later in the Carboniferous. This is consistent with the age of orogeny in the Urals. The second set of discrepancies may be explained by supposing that in Permian time north China and the Far Eastern Maritime Province lay well to the south and east. Then, before the late Mesozoic, it moved toward the Siberian Platform and became welded to it along the eastern Asiatic orogens.

Of course no firm case can be made on the basis of present data, but present evidence suggests the possibility that drift is a recurrent phenomenon and was not confined to a relatively late stage in earth history. Drift need not have been a process of increasing fragmentation of one or two primeval supercontinents, but rather a process of fragmentation followed by regrouping into new composite continents and by further fragmentation and regrouping episodes.

On the cause of long-period geomagnetic field changes. The suggestion is made that the cause of reversals may be in the mantle.

The occurrence in the Carboniferous of Australia of a rapid latitude change and a change in reversal frequency during a time of orogeny and marine regression suggests that there is a relation between events in the core and on the earth's surface. Reversals in the late Cenozoic rock formations are commonly said to be

'exact,' but the statistical errors are such that misalignments of several degrees would go unnoticed. *Khramov* [1958] has actually suggested, from work in Turkmenia, that axial directions of adjacent polarity epochs are misaligned by several degrees. He generalizes his opinion in the statement (p. 127) 'our data show that reversals occurred more often the quicker the coordinates of the pole change,' that is, that there is a relation between the rate of polar change and the frequency of reversal. It is possible, therefore, that the difference between the reversal at the Paterson transition and the usual Cenozoic reversal is one of degree and is not fundamental.

Surface changes can hardly respond to core motions (or vice versa) with the rapidity observed, and one is led to search for a common cause in the mantle. Let us assume that some redistribution of mass occurs there. This could, for example, be due to convection, and it is immaterial for these arguments whether it is general or localized upper mantle convection. Suppose this to be responsible, on the one hand, for episodic continental drift and tectonism, and, on the other hand, for polar wandering. The fluid parts of the earth are likely to be affected immediately by polar wandering. Oceanic water is likely to be redistributed, causing changes in sea level, and changes in the earth's core may alter the geomagnetic field. It is generally thought that very small changes in the motion of the turbulent fluid in the core could cause reversals, and polar wandering may bring these about; as the earth turns through its axis of rotation, the core is not expected to move as a whole but may, in response to the changed axis of rotation, alter its internal motions sufficiently to cause reversal. On this hypothesis, reversal frequency is a measure of tectonic activity.

The idea can be tested by observation. If it is true, the field can be expected to reverse frequently during times of active convection and tectonism. This appears to have occurred in the late Cenozoic when observed reversals are common, when the geological evidence suggests much orogenic activity, and when drift (shown by the disparity between Lower Tertiary and older poles from different places) and axial field changes (shown by the polar variation within the Cenozoic observed from the same place—France, for example) occurred. On the other hand, the field would, on this hypothesis, remain constant in direction and polarity during times of inactive convection and tectonism. This may have been so in the late Paleozoic, when, after a rapid field change correlated with the Hercynian orogeny and the mid-Carboniferous drift episode, there followed a time (about 50 m.y.) of constant polarity and remarkably uniform direction. There are already indications in the literature of a systematically reversed field of steady direction for part of the early Tertiary. It would be interesting to see whether this was prefaced, like the Kiaman magnetic interval, by a time of rapid polar change and, if so, how this change was related to the Mesozoic-Cenozoic boundary and the Laramide orogeny.

Many other results in the literature are indicative of a relation between reversal frequency and polar change. For example, at the bottom of the Torridonian Sandstone Series (Precambrian) [*Irving and Runcorn,* 1957] the polarity is uniform and the directions (or poles) are well grouped ($k = 40$). Within 30 m the directions change by 78° to a new axis about which reversals occurred frequently and about which the directions (irrespective of sign) are more scattered ($k = 12$). The axis change corresponds to a polar change of 46°, and it occurred 'rapidly compared with the time of duration of the stable directions' [*Irving,* 1957]; it marks a sudden increase in reversal frequency and in direction scatter.

Reversal and organic evolution. Uffen [1963, 1964] and *Simpson* [1966] describe a positive correlation between evolution and reversals. They suggest that this may be due to an increase in incoming radiation caused by withdrawal of the shielding dipole field during reversal and consequent increase of the mutation rate, so that polarity changes may be a major cause of evolution. Alternatively the apparent correlation can be explained in terms of the ideas mooted above.

It is probable that Uffen and Simpson themselves would agree that the correlations are imperfect, but they are perhaps as good as could be expected in view of incompleteness of the paleomagnetic record compared with fossil documentation. The imperfect nature of the correlations can be illustrated by an example.

The reversal at the end of the Kiaman magnetic interval occurs toward the end of the Paleozoic, and of all known reversals this should best show their effect on evolution because the polarity seems to have been constant for about 50 m.y. before that time. If evolutionary changes were predominantly due to dipole field withdrawal, they would commence suddenly—either at the boundary or shortly afterward, but not before. In fact the fossil changes are distributed about the boundary; for example, major changes in plants occur within the Permian (where there are no reversals)—before the boundary—and changes in amphibians at the end of the Triassic.

In recent work *Opdyke et al.* [1966] have found nine reversals in long ocean cores which contain four faunal zones (defined on Radiolarian extinctions). Two of the three faunal boundaries correspond roughly (to within 30 cm) to reversals but the other does not, and there are seven reversals of no observable paleontological significance. Again, correlations do not always occur, and when they do they are not exact.

Geologists have often thought that paleogeography (latitude and climate changes, for instance) is one of the major factors controlling evolution. This may still be true, and the sporadic correlation of evolution and reversals may arise from the common origin of both. If, on the one hand, motions in the mantle control outer tectonics and paleogeography and, on the other, produce polar wandering which yields reversals, an imperfect correlation between organic evolution and reversal such as are observed would be expected.

Note on amendment of earlier data. Carboniferous results from Australia based on observations of NRM without cleaning have been published earlier [*Cox and Doell,* 1960, reference numbers E50 and E51; *Irving,* 1960, reference numbers 110 and 111] and the results herein supersede them. The negative direction recorded earlier from the Upper Kuttung sediments are from the sediments *beneath* the Paterson toscanite (site H7(S)) which are shown herein to be unstable.

Appendix

Where possible, grid references on the 1 mile to the inch Military Map Series are given, the sheet name being given in brackets. Dip directions are given east of true north. Reference to geological descriptions are given if available.

Lower Kuttung Volcanics

H1. Martin's Creek andesite, two quarries: 32°34'S, 156°36'E; 640 742 and 640 750 (Paterson); flaggy jointing dipping 25° at 264°; *Osborne* [1925, p. 121].

H2. Horneblende andesite, road-cutting beneath Hudson's Peak: 32°38'S, 151°29'E; 497 652 (Singleton); flaggy jointing dipping 38° at 48°; second (from bottom) level in the stratigraphic list of *Browne* [1927, p. 218].

H3. Horneblende andesite, road cutting Lamb's Creek road: 32°36'S, 151°27'E; 459 696 (Singleton); flaggy jointing dipping 40° at 46°; sixth (from bottom) level in stratigraphic list of *Browne* [1927, p. 218].

H4. Toscanite, roadcut at The Gap ⅛ mile north of Melville Hill: 32°41'S, 151°31'E; 537 608 (Paterson); seventh (from bottom) level in the stratigraphic list of *Browne* [1927, p. 218].

H5. Toscanite, roadcut at 'Eelah corner' 1 mile west of Melville Hill: 32°41'S, 151°32'E; 547 605 (Paterson); eighth level (from bottom) in stratigraphic list of *Browne* [1927, p. 218].

H6. Felsite, roadcut at 'Eelah corner' 1 mile west of Melville Hill: 32°41'S, 151°32'E; 547 607 (Paterson); tenth (from bottom) level in the list of *Browne* [1927, p. 218].

Paterson Toscanite [*Osborne,* 1925, p. 130]

H7(L). Old quarry on roadside near Burnbrae on the William's River: 32°40'S, 151°45'E; 765 622 (Paterson); underlying sediments dipping 15° at 187°.

H8. Old quarry on hill above Paterson: 32°36'S, 151°37'E; 629 695 (Paterson); average bedding in adjacent sediments above and below lava dips 21° at 212°.

H9. Roadcut at Paterson: 32°36'S, 151°37'E; 635 691 (Paterson); bedding in sediments above and below lava dips 21° at 212°.

H10(L). Railway cut just southwest of railway station: 32°16'S, 151°37'E; 630 691 (Paterson); bedding in sediments above and below lava dipping 21° at 212°. It is here that the change called the *Paterson transition* occurs.

Upper Kuttung Bedded Rocks, Hunter Valley [*Osborne,* 1922-1925; *David,* 1950]

H7(S). Red tuffaceous very fine sandstone from bed 0.3 m thick beneath Paterson toscanite at H7(L).

H10(S). Pink and gray very fine sandstones overlying Paterson toscanite: beds dip 21° at 212°; see H10(L).

H11. Seaham varves, buff-colored siltstones from two roadcuts at the famous Seaham locality: 32°40'S, 151°44'E; 747 607 and 746 617 (Paterson); beds dip 18° at 204°.

H12. Pink and red claystones from Eagleton

quarry: 32°42'S; 151°45'E; 776 570 (Paterson); beds dip 84° at 284°.

H13(S). Pink and gray siltstones and very fine sandstones from New Council quarry: 32°40'E, 151°48'E; 819 751 (Paterson); beds dip 38° at 138°.

H14. Buff and pinkish siltstones from old quarry: 32°41'S; 151°48'E; 838 594 (Paterson); beds dip 48° at 170°.

Currabubulla Formation [*Carey*, 1937; *Voisey and Williams*, 1964; *Whetten*, 1965], Creek section at 31°16'S, 150°44'E

C1. Green fine sandstone dipping 36° at 251°.

C2. Green very fine sandstone dipping 36° at 251°.

C3. Green fine sandstone dipping 29° at 253°.

C4. Brown sandstone from prominent hard band in fissile shale, dipping 36° at 253°.

C5. Massive gray ignimbrite with pinkish bands dipping 37° at 247°.

Rocky Creek Conglomerate [*McKelvey and White*, 1964], Creek section on axis of syncline at 30°03'S, 150°17'E

RC1. Pink tuffaceous claystone from mouth of Hell Hole Creek, dipping 15° at 98°.

RC2. Red tuffaceous siltstone, dipping 15° at 344°.

RC3. Gray siltstone, dipping 8° at 285°.

RC4. Green siltstone, dipping 6° at 238°.

RC5. Purple siltstone, dipping 8° at 224°.

RC6. Pink tuffaceous siltstone, dipping 22° at 325°.

RC7. Lower sampling level of gray varves, dipping 17° at 245°. Samples consist of alternating layers of fine sandstone and claystone of variable thickness averaging about 0.5 cm. Pebbles up to 1.0 cm in size occur, and one (Figure 16) exceeded 4 cm.

RC8. Upper sampling level of gray varves, dipping 18° at 259°. Eight meters stratigraphically above RC8. Rock is predominantly of coarse sandstone grade with subordinate wisps of claystone, usually only 1 mm or so in thickness. Pebbles of 1-cm size are common.

Dikes

H13(D). Basalt dike: see H13(S).

M. Moonbi lamprophyre: 31°00'S, 151°04'E.

Goonoo Goonoo Mudstone of Parry Group

B1. Pyramid Hill arenite: level Clp_5 of *Crook* [1961*b*, p. 194]; fresh greenish sandstone from Flaggy Creek; 31°28'S, 150°52'E; 126 057 on sheet no. 3 'Nundle' in *Crook* [1961*b*]; beds dip 20° at 273°.

B2. Pyramid Hill arenite: level Clp_5 of *Crook* [1961*b*, p. 194]; fresh green sandstone from road-cut; 31°27'S, 151°01'E; 101 111 on sheet no. 3 'Nundle' in *Crook* [1961*b*]; beds horizontal.

B3. Well-bedded greenish-brown sandstone from Scrub Mountain conglomerate of *Crook* [1961*b*]: 31°17 S, 150°54 E; 238 082 on sheet no. 2 'Goonoo Goonoo' of *Crook* [1961*a*]; beds dip 20° at 275°.

B4. Greenish-brown sandstone from Garoo conglomerate of *Crook* [1961*b*]: 31°17'S, 150°53'E; 239 076 on sheet no. 2 'Goonoo Goonoo' of *Crook* [1961*a*]; beds dip 26° at 263°.

Acknowledgments. I am very grateful indeed to Professor J. C. Jaeger of the Australian National University for much encouragement and to Dr. B. C. Browne of Edgecliffe Sydney who was my instructor and guide in the field. I should like also to record my thanks to many other Australian geologists for help in field work and to Martin Ward and Alan Major for assistance with the measurements and analysis. Finally, I should like to thank Dr. E. R. Niblett of the Dominion Observatory and Dr. D. J. McLaren of the Geological Survey of Canada for many fruitful discussions of the puzzling correlations arising from these observations.

References

Briden, J. C., Palaeolatitudes and palaeomagnetic studies, Thesis, Australian National University, Canberra, 1964.

Briden, J. C., Ancient secondary magnetizations in rocks, *J. Geophys Res., 70,* 5205–5221, 1965.

Browne, W. R., The geology of the Gosforth district, N.S.W., *J. and Proc. Roy. Soc. N.S. Wales, 60,* 213–277, 1927.

Carey, S. W., The Carboniferous sequence in the Werrie Basin, *Proc. Linnean Soc. N.S. Wales, 59,* 351–379, 1937.

Carey, S. W., The asymmetry of the earth, *Australian J. Sci., 25,* 369–384, 1963.

Chamalaun, F. H., and K. M. Creer, Thermal demagnetization studies on the Old Red Sandstone of the Anglo-Welsh Cuvette, *J. Geophys. Res., 69,* 1607–1616, 1964.

Clegg, J. A., E. R. Deutsch, and D. H. Griffiths, Rock magnetism in India, *Phil. Mag., 1,* 419–431, 1956.

Cox, A. V., Analysis of the present geomagnetic field for comparison with paleomagnetic results, *J. Geomagnet., Geoelec., 13,* 101–112, 1962.

Cox, A. V., and R. R. Doell, Review of paleomagnetism, *Bull. Geol. Soc. Am., 71,* 645–768, 1960.

Creer, K. M., An analysis of the geomagnetic field using palaeomagnetic methods, *J. Geomagnet. Geoelec., 13,* 113–119, 1962.

Creer, K. M., Palaeomagnetic data from the Gondwanic continents, *Phil. Trans. Roy. Soc. London, A, 258,* 27–40, 1965.

Creer, K. M., E. Irving, and S. K. Runcorn, Geophysical interpretation of palaeomagnetic directions from Great Britain, *Phil. Trans. Roy. Soc. London, A, 250,* 144–156, 1957.

Creer, K. M., A. E. M. Nairn, and E. Irving, Palaeomagnetism of the Great Whin Sill, *Geophys. J. 2,* 306–323, 1959.

Crook, K. A. W., Stratigraphy of the Tamworth group, *J. and Proc. Roy. Soc. N.S. Wales, 94,* 173–188, 1961*a*.

Crook, K. A. W., Stratigraphy of the Parry group, *J. and Proc. Roy. Soc. N.S. Wales, 94,* 189–207, 1961*b*.

David, T. W. E. (edited by W. R. Browne), *The geology of the Commonwealth of Australia,* 747 pp., Arnold, London, 1950.

Deutsch, E. R., Recent palaeomagnetic evidence for the northward movement of India, *J. Alberta Soc. Petrol. Geologists., 6,* 155–162, 1958.

Du Toit, A. L., *Our Wandering Continents,* 336 pp., Oliver and Boyd, Edinburgh, 1937.

Everdingen, R. O., van, Studies on the igneous rock complex of the Oslo region, *Skrift. Norske Videnskaps-Akad. Olso, Mat.-Naturv., 1,* 1–80, 1960.

Evernden, J. F., and J. R. Richards, Potassium-argon ages in eastern Australia, *J. Geol. Soc. Australia, 9,* 1–50, 1962.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London, A, 217,* 295–305, 1953.

Furon, R., *Geology of Africa,* 377 pp., Oliver and Boyd, Edinburgh, 1963.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Res., 54,* 131–167, 1949.

Green, R., Palaeomagnetism of Devonian and Tertiary rocks from Australia, Ph.D. thesis, Australian National University, Canberra, 1959.

Green, R., Palaeomagnetism of some Devonian rock formations in Australia, *Tellus, 13,* 119–124, 1961.

Haughton, S. H., *Stratigraphic Geology of Africa,* 365 pp., Oliver and Boyd, Edinburgh, 1963.

Helsley, C. E., Paleomagnetic results from the Lower Permian Dunkard series of West Virginia, *J. Geophys. Res., 70,* 413–424, 1965.

Irving, E., Directions of magnetization in the Carboniferous glacial varves of Australia, *Nature, 180,* 280–281, 1957.

Irving, E., Rock magnetism: a new approach to some palaeogeographic problems, *Phil. Mag. Suppl. Adv. Phys., 6,* 194–218, 1957.

Irving, E., Palaeomagnetic directions and pole positions, *Geophys. J., 3,* 96–111, 1960.

Irving, E., Paleomagnetism of the Narrabeen Chocolate Shales and the Tasmanian dolerite, *J. Geophys. Res., 68,* 2283–2287, 1963.

Irving, E., *Paleomagnetism,* 399 pp., John Wiley & Sons, New York, 1964.

Irving, E., and R. Green, Polar movement relative to Australia, *Geophys. J., 1,* 64–71, 1958.

Irving, E., and L. G. Parry, The magnetism of some Permian rocks from New South Wales, *Geophys. J., 7,* 395–411, 1963.

Irving, E., W. A. Robertson, P. M. Stott, D. H. Tarling, and M. A. Ward, Treatment of partially stable sedimentary rocks showing planar distribution of directions of magnetization, *J. Geophys. Res., 66,* 1927–1933, 1961*a*.

Irving, E., and S. K. Runcorn, Analysis of the palaeomagnetism of the Torridonian sandstone series of northwest Scotland, *Phil. Trans. Roy. Soc. London, A, 250,* 83–99, 1957.

Irving, E., P. M. Stott, and M. A. Ward, Demagnetization of igneous rocks by alternating magnetic fields, *Phil. Mag., 6,* 225–241, 1961*b*.

Khramov, A. N., *Palaeomagnetism and Stratigraphic Correlation,* 218 pp., Gostoptechizdat, Leningrad, 1958. English translation published by Australian National University, Canberra, 1960.

Khramov, A. N., V. P. Rodionov, and R. A. Komissarova, New data on the Palaeozoic history of the geomagnetic field in the U.S.S.R., *Present and Past Geomagnetic Field,* pp. 206–213, Nauka Press, Moscow, 1965.

King, L. C., *The Morphology of the Earth,* 699 pp., Oliver and Boyd, Edinburgh, 1962.

King, R. F., The remanent magnetism of artificially deposited sediments, *Monthly Notices Roy. Astron. Soc., Geophys. Supp., 7,* 115–134, 1955.

Köppen, W., and A. Wegener, *Die Klimate der geolischen Vorzeit,* Bornträger, Berlin, 1924.

Kulp, J. L., Geologic time scale, *Science, 133,* 1105–1114, 1961.

Maack, R., Zur paläogeographie des Gondwanalandes, *Intern. Geol. Congr., 21st, Copenhagen,* pt. 12, 35–55, 1960.

Manwaring, E. A., Palaeomagnetism of some igneous rocks of the Sydney basin, New South Wales, *Proc. Roy. Soc. N. S. Wales, 96,* 141–151, 1963.

Mercanton, P. L., Inversion de l'inclinaison magnétique terrestre aux âges geologiques, *Terrest. Magnetism Atmospheric Elec., 31,* 187–190, 1926.

McElhinney, M. W., Statistical significance of the fold test in palaeomagnetism, *Geophys. J., 8,* 338–340, 1964.

McKelvey, B. C., and A. H. White, *Geological map of New England, Horton sheet 290,* University of New England, Armidale, N.S. Wales, 1964.

Mumme, W. G., A note on the mixed polarity of magnetization in Cainozoic basalts in Victoria, Australia, *Geophys. J., 6,* 546–549, 1962.

Opdyke, N. D., B. Glass, J. D. Hays, and J. Foster, A paleomagnetic study of Antarctic deep sea cores, *Science,* in press, 1966.

Osborne, G. D., Geology and petrography of the Clarencetown–Paterson District, Parts 1-4, *Proc. Linnean Soc. N.S. Wales, 47,* 161–198, *47,* 519–534, *50,* 67–79, *50,* 112–138, 1922 and 1925.

Rees, A. I., The effect of water currents on the magnetic remanence and anisotropy of susceptibility of some sediments, *Geophys. J., 5,* 235–251, 1961.

Robertson, W. A., A palaeomagnetic study of igneous rocks from eastern Australia, Thesis, Australian National University, Canberra, 1962.

Robertson, W. A., The paleomagnetism of some Mesozoic intrusives and tuffs from eastern Aus-

tralia, *J. Geophys. Res., 68,* 2299–2312, 1963.
Robertson, W. A., Palaeomagnetism of the monzonite porphyry from Milton, New South Wales, *Pure Appl. Geophys., 59,* 93–99, 1964.
Simpson, J. F., Evolutionary pulsations and geomagnetic polarity, *Bull. Geol. Soc. Am., 77,* 197–204, 1966.
Stubbs, P. H. S., A palaeomagnetic study of the British and European Triassic and of the British Old Red Sandstone, Thesis, University of London, 1958.
Teichert, C., Fossile Riffe als Klimazeugen in Australien, *Geol. Rundschau, 40,* 33–38, 1952.
Uffen, R. J., Influence of the earth's core on the origin and evolution of life, *Nature, 198,* 143–144, 1963.
Uffen, R. J., The evolution of the interior of the earth and its effects on biological evolution, *Intern. Geol. Congr., 22nd, India,* 1964.
Vincenz, S. A., and J. McG. Bruckshaw, Note on the probability distribution of a small number of vectors, *Proc. Cambridge Phil. Soc., 56,* 21–26, 1960.
Vlassov, A. Ya., and G. V. Kovalenko, The history of the geomagnetic field, edited by A. C. Kalashnikov, *Bull. Acad. Sci. USSR, Geophys. Ser., English Transl., no. 9,* 1243–1279, 1961.
Vlassov, A. Ya., and P. V. Popova, Position of the pole in Permian, Triassic and Cretaceous periods according to findings of paleomagnetic studies of sedimentary rocks in the Soviet Far Eastern Maritime Province, in *Rock Magnetism and Palaeomagnetism.,* pp. 333–339, Siberian Academy of Sciences, Krasnoyarsk, 1963.
Vlassov, A. Ya., A. V. Popova, A. G. Zvegintsev, and E. K. Rodicheva, Paleomagnetic investigations of Devonian sedimentary strata in the central part of the Krasnoyarsk district, *Bull. Acad. Sci. USSR, Geophys. Ser., English Transl.,* no. 7, 1022–1024, 1961.
Voisey, A. H., and K. L. Williams, The geology of the Carroll-Keepit–Rangari area of New South Wales, *J. and Proc. Roy. Soc. N.S. Wales, 97,* 65–72, 1964.
Wang, T., H. Teng, C. Li, and S. Yeh, *Acta Geophys. Sinica, 9,* 125–138, 1960.
Watson, G. S., and E. Irving, Statistical methods in rock magnetism, *Monthly Notices Roy. Astron. Soc., Geophys. Suppl., 7,* 289–300, 1957.
Whetten, J. T., Carboniferous glacial rocks from the Werrie basin, N.S.W., Australia, *Bull. Geol. Soc. Am., 76,* 43–54, 1965.
Wilson, R. L., and C. W. F. Everitt, Thermal demagnetization of some Carboniferous lavas for palaeomagnetic purposes, *Geophys. J., 8,* 149–164, 1963.

(Received June 8, 1966;
revised September 6, 1966.)